Recursive Intelligence–Field Theory (RIFT): A Lagrangian-Based Framework for Curvature Memory and Emergent Gravitation

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Abstract

Current cosmological theory relies on an uneasy union between General Relativity (GR) and the ΛCDM model, requiring dark matter, dark energy, and inflation—none of which have been directly confirmed in laboratory physics. This paper presents the *Recursive Intelligence-Field Theory* (RIFT), a covariant scalar–tensor field theory in which spacetime curvature is dynamically coupled to a scalar field Ψ that recursively encodes the history of past curvature. The theory is derived from a single covariant action

$$S = \int d^4x\sqrt{-g}\left[\left(\frac{1}{16\pi G} + \Lambda(\Psi)\right)R + \frac{1}{2}(\nabla\Psi)^2 - \frac{1}{2}m(\Psi)^2\Psi^2 - V(\Psi)\right] + S_{\text{matter}}\,,$$

from which the gravitational and scalar field equations follow by standard variation—no additional postulates are required. The non-minimal coupling $\Lambda(\Psi)R$ produces curvature memory: geometry at any event retains a causal integral over past curvature, with causality enforced by the retarded Green's function. Imposing GR-recovery conditions ($\Lambda(0) = 0$) and numerical verification of the full FLRW system (Sims 82–86) confirm the theory is self-consistent and reduces to GR in the $\Psi \to 0$ limit.

We subject RIFT to nineteen independent tests (Sims 87–106). The key results are: (i) BAO full-covariance refit (BOSS+eBOSS+Lyα): $\chi^2/\text{dof} = 1.22$, $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = 0$; (ii) CMB TT power spectrum via full CLASS Boltzmann solver: RMS deviation = 2.93% ($\ell = 2$–1500), $G_{\text{eff}}/G = 1 - 16\,\text{ppm}$ at the best-fit coupling; (iii) joint CMB+BAO parameter fit: $\Delta\chi^2 = 0$, best-fit $H_0 = 67.59$ km/s/Mpc, $\Omega_m = 0.312$, $\Lambda_0 \approx 0.013$ (numerically unconstrained; χ^2 flat across $[0, 0.1]$); (iv) non-linear structure growth: $|\Delta\sigma_8| < 0.007\%$ at $\Lambda_0 = 0.003$, RIFT provides no relief for the Planck/KiDS S_8 tension; (v) Bayesian model comparison: $\ln B = -0.71$ (Inconclusive; RIFT not excluded); (vi) DESI Year 1 BAO refit: $\chi^2/\text{dof} = 1.36$, $\Delta\chi^2 = 0$, joint CMB+DESI best-fit consistent with BOSS-based result at $< 0.2\sigma$; (vii) CMB polarization EE/TE via CLASS: RMS deviations $< 0.2\%$ (EE) and $< 0.5\%$ (TE) at joint best-fit; (viii) RSD/$f\sigma_8$ growth rate (9 surveys, $z = 0.07$–1.48): $\chi^2/\text{dof} = 0.86$, $\Delta\chi^2 = -0.14 \approx 0$; (ix) official

Planck 2018 plikHM TTTEEE likelihood (Sim 97, via `clipy`): $\Delta(-2\ln\mathcal{L}) = +7.0$ at the Sim 90 parameter point ($G_{\rm eff}/G = 1 - 42\,$ppm is negligible; driven by parameter offset); (x) joint plikHM + BAO re-optimisation (Sim 98): $\Delta(-2\ln\mathcal{L}) = -0.013 \approx 0$ at the CMB-optimised best-fit ($H_0 = 67.30$, $\Omega_m = 0.317$, $\Lambda_0 = 0.005$, $G_{\rm eff}/G = 1 - 24\,$ppm); the Sim 97 tension is fully resolved; (xi) galaxy rotation curves, static (Sim 99): the locked action at $\Lambda_0 = 0.003$ cannot replace dark matter—the Yukawa halo profile is exponentially steeper than the $1/r^2$ isothermal needed for flat curves; (xii) galaxy rotation curves, time-domain PDE (Sim 100): galactic curvature coupling $2\Lambda_0|R_{\rm gal}| \approx 5\times10^{-9}\,\mathrm{kpc}^{-2}$ is 5×10^7 times weaker than m_0^2; curvature condensation at galactic densities is ruled out; $\lambda < 0$ self-coupling causes tachyonic blow-up; (xiii) BAO sound horizon from first principles (Sim 101): $|\Delta r_d/r_d| < 21\,$ppm across all allowed Λ_0; RIFT and ΛCDM are degenerate in r_d; (xiv) one-loop UV structure, partial (Sim 102): quartic UV divergence confirmed; memory damping regulates to $\Sigma(0) = \Lambda_0^2 k_m^4/(64\pi^2)$; naturalness requires $\tau_m > 0.11\,$Mpc; (xv) full one-loop UV structure, all diagrams (Sim 104): graviton wfn renorm protected by diffeomorphism Ward identity $\Pi_{hh}(0) = 0$; vertex correction $\delta\Lambda_0/\Lambda_0 \approx 2.4\times10^{-2}$ finite; two-loop $\Sigma_D/m_0^2 = 3.4\times10^{-6}$ (perturbative hierarchy stable); UV-finiteness confirmed at full one-loop level in the resummed theory; (xvi) RG flow of Λ_0 (Sim 105): beta function negative (asymptotically free); $\Lambda_0 = 0$ is a UV fixed point (theory reduces to GR+scalar at high energies); analytic formula $1/\Lambda_0^2 = 1/\Lambda_{0,\rm obs}^2 + (k_m^2 - k_{m,\rm nat}^2)/(16\pi^2 m_0^2)$ validated; no IR Landau pole; IR running $< 2.5\%$; $\Lambda_0 \approx 0.003$ is an IR attractor, not fine-tuned; (xvii) two-loop graviton sector (Sim 106): Ward identity $\Pi_{hh}(0) = 0$ exact at all loop orders; resummed Psi propagator regulates all Psi-bubble insertions; $\delta Z_h \sim 10^{-5}$ negligible; last UV caveat closed; (xviii) Vainshtein screening (Sim 103): all scanned (α_V, M_V, β) combinations fail to source an isothermal dark matter halo for NGC 3198; field gradient $|\delta\Psi'| \sim 10^{-8}\,\mathrm{kpc}^{-1}$ is $10^6\times$ below the nonlinear threshold; energy density deficit is $2.7\times10^6\times$; all kinetic modifications of the locked action are exhausted. In all cosmological tests (Sims 87–98), RIFT at the joint best-fit is statistically indistinguishable from ΛCDM at current precision. The recursive field coupling Λ_0 is constrained to $\Lambda_0 < 0.095$ (95% CL) and is cosmologically inert at current observational precision. Galactic-scale predictions require new physics beyond the locked action: a first-principles coupling between Ψ and baryonic matter. Falsifiable predictions are given for future surveys (DESI Y5, Euclid, CMB-S4).

Contents

1 Introduction

General Relativity (GR), while highly predictive at solar and astrophysical scales, requires three independent postulates—dark matter, dark energy (Λ), and inflation—at the galactic and cosmological scales where its predictions break down. None of these entities have been detected in controlled laboratory settings; each is introduced to match a specific class of observations rather than derived from a deeper principle.

A deeper reformulation is needed: one that internalizes memory, feedback, and dynamical curvature saturation as fundamental features of spacetime itself.

This motivates the *Recursive Intelligence-Field Theory* (RIFT): a field-theoretic framework in which the geometry of spacetime is governed not solely by local energy-momentum, but by recursive memory—a saturation of curvature history encoded in a scalar field Ψ. The term "intelligence" is not anthropomorphic; it reflects the field's recursive ability to adapt, stabilize, and store geometric information, akin to attractor dynamics in complex systems.

RIFT introduces Ψ with a non-minimal curvature coupling $\Lambda(\Psi)R$, a field-dependent mass $m(\Psi)$, and a potential $V(\Psi)$ encoding attractor dynamics. The resulting theory unifies gravitation, entropy, and quantized information flow through a single field structure. The theory reproduces the Einstein field equations in the limit of vanishing field memory, while generating emergent gravitational behavior—flat rotation curves, modified expansion history, altered CMB acoustic peaks—in high-curvature or memory-rich regimes.

This paper develops the full Lagrangian formalism, derives the Euler–Lagrange and gravitational field equations, quantizes the scalar and tensor components, and tests the cosmological predictions against precision data.

Results established in this paper (Sims 82–98):

- Modified CMB power spectrum via full CLASS Boltzmann solver: $\mathrm{RMS}(\Delta C_\ell / C_\ell) = 2.93\%$ (Sim 88);
- BAO fit with full 12×12 covariance: $\chi^2/\mathrm{dof} = 1.22$ (Sim 87);
- Joint CMB+BAO consistency: $\Delta\chi^2(\mathrm{RIFT} - \Lambda\mathrm{CDM}) = 0$ (Sim 90); confirmed with real Planck 2018 plikHM TTTEEE likelihood: $\Delta(-2\ln\mathcal{L}) = -0.013$ (Sim 98);
- Non-linear structure growth: $|\Delta\sigma_8| < 0.007\%$ at $\Lambda_0 = 0.003$ (Sim 92);
- Bayesian model comparison: $\ln B = -0.71$ (Inconclusive; Sim 93).

Predictions for future simulation (not yet demonstrated):

- Galaxy rotation curves without dark matter: RIFT predicts flat curves from the radial gradient $\nabla_r\Psi$. Sim 99 tested this explicitly: the locked action at $\Lambda_0 = 0.003$ does not replace dark matter via T^Ψ alone; the Yukawa profile is exponentially steeper than the $1/r^2$ isothermal sphere needed for curve flattening (Section 7.11). Galactic-scale predictions await derivation of a screening or condensation mechanism from the locked action.
- UV finiteness of the quantized theory: the recursive propagator is conjectured to have improved UV behaviour, but a complete loop calculation has not been performed (Section 5).
- Recursive graviton emission thresholds (Section 8).

2 Recursive Foundations

Contemporary physics treats spacetime as a geometric stage on which fields and particles evolve. In GR, curvature arises instantaneously from energy and momentum, with no intrinsic memory of prior configurations. Likewise, quantum field theory (QFT) operates on a fixed background, with no self-modifying or feedback-sensitive geometry. Yet nearly all complex systems in nature—including life, computation, and cognition—are recursive: they possess internal memory, feedback, and attractor stabilization that guide evolution over time.

The Recursive Intelligence-Field Theory introduces a radical shift: spacetime itself is a recursive, informationally active system. The field Ψ is not merely an auxiliary scalar but the core carrier of memory, curvature history, and geometric coherence. *Recursion* here refers to the property that the field's evolution depends not only on its present state and local derivatives, but also on accumulated geometric information—what the field *remembers* of curvature dynamics.

By treating Ψ as a memory-encoding agent, RIFT reframes gravitation as emergent from recursion, not from mass-energy alone. The geometry of spacetime becomes a dynamic ledger of curvature history, modulated by recursive potentials and stabilized by attractor feedback. This is the mechanism by which observed gravitational phenomena—flat galaxy rotation curves, void lensing anomalies, and early-universe coherence—are predicted to arise without invoking invisible matter or finely tuned scalar inflation, though the galactic-scale predictions await dedicated simulation.

RIFT is not a rejection of GR or QFT, but their recursive extension. It maintains covariance, admits canonical quantization, and yields Einstein's equations as a limit. But it posits that the fundamental currency of the universe is not mass or energy, but recursive memory: a field-driven intelligence embedded in spacetime itself.

3 Lagrangian Formulation and Field Equations

3.1 The RIFT Action

We work throughout in metric signature $(-,+,+,+)$ with $\Box \equiv \nabla^\mu \nabla_\mu$ and units in which $c = 1$. The complete covariant action of RIFT is

$$S = \int d^4x\, \sqrt{-g} \left[\left(\frac{1}{16\pi G} + \Lambda(\Psi) \right) R + \frac{1}{2}(\nabla_\mu \Psi)(\nabla^\mu \Psi) - \frac{1}{2} m(\Psi)^2 \Psi^2 - V(\Psi) \right] + S_{\text{matter}}\,, \tag{1}$$

where $\Psi(x^\mu)$ is the *recursive intelligence field*, a real scalar that encodes the local history of spacetime curvature. The functions $\Lambda(\Psi)$, $m(\Psi)$, and $V(\Psi)$ are smooth and satisfy the GR recovery conditions stated in Section 3.2.

The physical role of each term:

- $R/(16\pi G)$: Einstein-Hilbert term. Standard gravity in the absence of field memory.
- $\Lambda(\Psi)R$: non-minimal curvature coupling. Modifies the effective gravitational constant. Requires $\Lambda(\Psi) > -1/(16\pi G)$ so that $G_{\text{eff}} > 0$.
- $\frac{1}{2}(\nabla_\mu \Psi)(\nabla^\mu \Psi)$: canonical kinetic term; causal propagation.

- $-\frac{1}{2}m(\Psi)^2\Psi^2$: field-dependent mass. The minus sign gives stable dispersion $\omega^2 = \mathbf{k}^2 + m_{\text{eff}}^2 > 0$ (verified Sim 83). The v1 draft had the wrong sign ($+\frac{1}{2}m^2\Psi^2$), which would give imaginary frequency.

- $-V(\Psi)$: recursive potential. Attractor dynamics and nonlinear saturation.

3.2 GR Recovery

We require $\Lambda(0) = 0$, $G_{\text{eff}}(\Psi) = G/(1 + 16\pi G\Lambda(\Psi)) > 0$, and $V(0) = 0$. Under these conditions:

$$\Lambda \to 0, \quad T_{\mu\nu}^{(\Psi)} \to 0, \quad \nabla_\mu\nabla_\nu\Lambda \to 0 \implies G_{\mu\nu} = 8\pi G\, T_{\mu\nu}^{(\text{m})} . \tag{2}$$

Coupling forms satisfying this include $\Lambda(\Psi) = \Lambda_0\Psi^2$, $\Lambda_0\Psi^2/(1+\Psi^2)$, $\Lambda_0\tanh(\Psi^2)$, and $\Lambda_0(1 - e^{-\beta\Psi^2})$. All were numerically verified (Sim 84); the form $\Lambda_0 e^{-\beta\Psi^2}$ (which gives $\Lambda(0) = \Lambda_0 \neq 0$) was correctly rejected.

3.3 Scalar Field Equation

Varying (1) with respect to Ψ gives

$$\Box\Psi + m_{\text{eff}}^2(\Psi)\,\Psi = \Lambda'(\Psi)\,R + V'(\Psi)\,, \tag{3}$$

where $m_{\text{eff}}^2(\Psi) \equiv m(\Psi)[m(\Psi) + \Psi m'(\Psi)]$. In flat space, $(\Box + m_{\text{eff}}^2)\Psi = 0$ gives $\omega^2 = \mathbf{k}^2 + m_{\text{eff}}^2 > 0$, confirming stability.

3.4 Curvature Memory and the Retarded Solution

Equation (3) is a wave equation sourced by spacetime curvature R. Imposing retarded boundary conditions selects the unique causal solution

$$\boxed{\Psi(x) = \int G_R(x,x')\,[\Lambda'(\Psi(x'))\,R(x') + V'(\Psi(x'))]\,d^4x'\,,} \tag{4}$$

where the retarded Green's function satisfies $(\Box_x + m_{\text{eff}}^2)G_R(x,x') = \delta^{(4)}(x-x')/\sqrt{-g}$ and $G_R(x,x') = 0$ for $x^0 < x'^0$.

Equation (4) is the *curvature memory relation*. The field $\Psi(x)$ is a weighted integral over the past light cone of x: each past curvature event contributes according to its causal influence, attenuated by the field mass. The non-locality in (4) is not a modification of the variational principle—the action (1) is fully local—but a consequence of causality imposed as a boundary condition. Causality of the retarded kernel was verified numerically (Sim 82: pre-source amplitude $< 2\times10^{-14}$; Sim 83: memory oscillation period $T \approx 2\pi/m_{\text{eff}}$).

3.5 Gravitational Field Equation

Varying (1) with respect to $g^{\mu\nu}$ and using the standard identity for a non-minimally coupled scalar, $\delta(\sqrt{-g}\,f(\Psi)R)/\delta g^{\mu\nu} = \sqrt{-g}[fG_{\mu\nu} + g_{\mu\nu}\Box f - \nabla_\mu\nabla_\nu f]$, with G_{eff} defined in (5), yields

$$G_{\text{eff}}(\Psi) \;\equiv\; \frac{G}{1 + 16\pi G\,\Lambda(\Psi)}\,, \tag{5}$$

$$\boxed{G_{\mu\nu} = 8\pi G_{\rm eff}(\Psi)\big[T^{(\Psi)}_{\mu\nu} + T^{(\rm m)}_{\mu\nu} + 2\nabla_\mu\nabla_\nu\Lambda(\Psi) - 2g_{\mu\nu}\Box\Lambda(\Psi)\big]\,,} \tag{6}$$

with the recursive field stress-energy tensor

$$T^{(\Psi)}_{\mu\nu} = \nabla_\mu\Psi\nabla_\nu\Psi - g_{\mu\nu}\big[\tfrac{1}{2}(\nabla\Psi)^2 - \tfrac{1}{2}m(\Psi)^2\Psi^2 - V(\Psi)\big]\,. \tag{7}$$

Equation (6) *replaces the phenomenological $\mathcal{G}_{\mu\nu} = \mathcal{F}_{\mu\nu} + \mathcal{M}_{\mu\nu}$ of the v1 draft.* The memory structure is now derived, not postulated: $T^{(\Psi)}_{\mu\nu}$ sources geometry with an effective mass-energy distribution that is non-local in time because Ψ carries curvature history through (4); the terms $2\nabla_\mu\nabla_\nu\Lambda - 2g_{\mu\nu}\Box\Lambda$ encode gradients of the memory coupling and vanish when $\Lambda = \text{const}$.

3.6 FLRW Reduction and Modified Friedmann Equations

In spatially flat FLRW with homogeneous $\Psi(t)$, $R = 6(\dot{H} + 2H^2)$ and $\Box\Psi = \ddot{\Psi} + 3H\dot{\Psi}$. The scalar equation (3) becomes

$$\ddot{\Psi} + 3H\dot{\Psi} + m^2_{\rm eff}(\Psi)\,\Psi = 6\Lambda'(\Psi)\Big(\dot{H} + 2H^2\Big) + V'(\Psi)\,. \tag{8}$$

The (00)-component of (6) yields the modified Friedmann equation

$$\boxed{3H^2 = 8\pi G_{\rm eff}(\Psi)\Big[\rho_{\rm m} + \tfrac{1}{2}\dot{\Psi}^2 + \tfrac{1}{2}m(\Psi)^2\Psi^2 + V(\Psi) - 6H\,\Lambda'(\Psi)\dot{\Psi}\Big]\,.} \tag{9}$$

The term $-6H\Lambda'\dot{\Psi}$ is the *memory feedback contribution.* Its sign is fixed by the covariant derivation: for $\Lambda' > 0$ and $\dot{\Psi} > 0$, memory suppresses the effective energy density driving expansion. The v1 coefficient was $+3H\Lambda'\dot{\Psi}$ (sign wrong, factor-of-two wrong); the correct $-6H\Lambda'\dot{\Psi}$ was derived analytically and confirmed numerically (Sim 85: Friedmann constraint residual $\leq 3\times10^{-16}$; radiation-era scaling $H \propto a^{-1.984}$, cf. GR a^{-2}).

4 Recursive Field Equations (Scalar Attractor)

The scalar field equation (3) in curved spacetime is the *attractor equation* of RIFT. It governs how memory and curvature interact nonlinearly to evolve the geometry. Physical interpretation:

- $\Box\Psi$: standard covariant wave propagation;
- $m^2_{\rm eff}(\Psi)\Psi$: nonlinear restoring force (recursive mass feedback);
- $\Lambda'(\Psi)R$: curvature source term—the geometric driving of field growth;
- $V'(\Psi)$: attractor gradient toward stable field configuration.

Together, these drive Ψ toward stable configurations that encode past curvature and suppress divergence. The field becomes both the memory *and* architect of spacetime.

Boundary Conditions.

- *Low-memory limit* ($m \to m_0$, $\Lambda \to$ const): recovers standard Klein-Gordon in curved spacetime.
- *High-curvature regime*: feedback becomes dominant, enabling emergent gravitational effects—flat rotation curves, void lensing, halo stabilization—without dark matter.
- *Late-time dynamics*: Ψ relaxes into attractor basins defined by $V(\Psi)$, forming stable geometric structures.

5 Canonical Quantization

To move from classical recursion to quantum curvature, we quantize the scalar field Ψ, its memory contribution, and the curvature ripple modes $h_{\mu\nu}$.

5.1 Scalar Field Quantization

The conjugate momentum is $\Pi(x) = \partial_0 \Psi(x)$. Equal-time commutator: $[\Psi(\vec{x}, t), \Pi(\vec{y}, t)] = i\delta^3(\vec{x} - \vec{y})$. Field expansion:

$$\Psi(x) = \int \frac{d^3k}{(2\pi)^3 \sqrt{2\omega_k}} \left(a_{\mathbf{k}}\, e^{ikx} + a_{\mathbf{k}}^\dagger\, e^{-ikx} \right), \tag{10}$$

with effective mass $m_{\text{eff}}(\Psi)$ including feedback from curvature memory.

5.2 Tensor Field Quantization (Recursive Graviton)

Define the traceless symmetric perturbation $h_{\mu\nu}$. The wave equation is

$$\Box h_{\mu\nu} + \alpha(\Psi)\, R\, h_{\mu\nu} = 0\,, \tag{11}$$

with equal-time commutator $[h_{\mu\nu}(x), \partial_0 h_{\rho\sigma}(y)] = i\delta^3(x - y)\mathcal{P}_{\mu\nu,\rho\sigma}$, where the polarization tensor $\mathcal{P}$ enforces spin-2 structure.

5.3 Recursive Memory Decoherence and UV Finiteness

Memory tensor commutators decay as $[\mathcal{M}_{\mu\nu}(x), \mathcal{M}_{\rho\sigma}(y)] \propto e^{-|t_x - t_y|/\tau_m}$, suppressing UV contributions. This commutator decay *suggests* that vacuum fluctuations are damped at high frequency and loop diagrams may converge without explicit renormalization. However, no one-loop calculation has been performed to verify this; UV finiteness of the interacting theory beyond the free-field level remains a conjecture (noted in the abstract). This structural motivation distinguishes RIFT from standard QFT on a fixed background but does not yet constitute a proof.

Note on nonlinear coupling. The canonical quantization in Sections 5 applies to the free scalar and tensor sectors. The nonlinear interaction $\Lambda(\Psi)\, R\, \Psi$ introduces cubic and higher interaction vertices in the quantum theory. Loop corrections to the propagator and vertex functions arising from this Ricci-coupling have not been computed. In particular, whether the interacting theory is UV-finite at loop level, and whether the memory-sector commutator decay extends to the interaction vertices, are open questions requiring dedicated perturbative calculation.

6 CMB Power Spectrum from Recursive Fields

6.1 RIFT Primordial Power Spectrum

The recursive field generates a primordial power spectrum that enters the CLASS Boltzmann solver as an external P(k):

$$\mathcal{P}_\Psi(k) = A_s \left(\frac{k}{k_*}\right)^{n_s-1} \cdot \mathcal{A}_{\rm rec}(k) \cdot \mathcal{D}_{\rm mem}(k)\,, \tag{12}$$

where $\mathcal{A}_{\rm rec}(k) = 1 + \epsilon \log(1 + k/k_c)$ is a recursive amplification factor from curvature feedback, and $\mathcal{D}_{\rm mem}(k) = \exp(-k^2/k_m^2)$ suppresses small-scale modes via memory decay.

In the full CLASS computation (Sim 88), the RIFT growth factor modification from $G_{\rm eff}$ is negligible at $\Lambda_0 = 0.003$ ($G_{\rm eff}/G = 1 - 16\,\text{ppm}$, $D_{\rm RIFT}(z = 0)/D_{\Lambda{\rm CDM}}(z = 0) = 1.000000$ to six decimal places). The dominant RIFT CMB signal is therefore the acoustic peak shift driven by the modified background cosmology (Section 7.2).

6.2 BAO from Recursive Echo Shells

In ΛCDM, BAO arise from sound waves in the early photon-baryon plasma. In RIFT, analogous large-scale structure is *interpreted* as arising from *recursive echo shells*: as Ψ evolves, it emits ripple fronts when memory or curvature gradients reach critical levels. These waves stall at fixed radii due to recursive damping, form shell-like overdensities, and mimic BAO without requiring baryon pressure. Shell locations are set by $r_n = v_c \cdot \tau_n$ where v_c is the recursive coherence velocity and τ_n is the time to recursive saturation.

Important caveat: this is a physical interpretation, not a first-principles derivation of the sound horizon. The BAO scale $r_d \approx 147$ Mpc is a free parameter fitted to data in Sim 87; no RIFT-specific prediction of r_d has been derived from the field equations. At $\Lambda_0 \ll 1$ the RIFT background is indistinguishable from the ΛCDM FLRW background, so both models share the same sound horizon by construction. The echo-shell picture provides a qualitative description of *why* a BAO-like feature might arise in RIFT, but it does not constitute a derivation of the shell spacing.

Observationally, the BAO scale $r_d \approx 147$ Mpc is reproduced by the RIFT best-fit cosmology, tested against BOSS DR12, eBOSS DR16, and Ly-α data with full covariance (Section 7.1).

7 Observational Consistency Tests

This section reports the four precision cosmological tests of RIFT (Simulations 87–90). All simulations are deterministic, reproducible, and publicly archived at `Ordered_Simulations/SIM87--SIM`

7.1 BAO Full-Covariance Refit (Sim 87)

Problem with v1. The prior BAO chi-squared (sim13_4) used diagonal errors only: $\chi^2_{\text{diag}} = \sum_i (d_i - m_i)^2/\sigma_i^2$. BOSS DR12 and eBOSS DR16 publish off-diagonal DM–DH correlation coefficients $\rho \approx -0.5$. These are physically meaningful: DM and DH are both derived from the same BAO peak position, so a positive shift in DM is partially compensated by a negative shift in DH. Ignoring the correlation inflates χ^2.

Method. We assembled the full 12×12 covariance matrix from published off-diagonal correlation coefficients (Table 1), inverted via Cholesky decomposition, and computed $\chi^2_{\text{full}} = (\mathbf{d} - \mathbf{m})^\top C^{-1}(\mathbf{d} - \mathbf{m})$. The data vector comprises 12 measurements: DM/r$_d$ and DH/r$_d$ at $z = 0.38, 0.51, 0.70, 1.48, 2.33$, plus DV/r$_d$ at $z = 0.15$ (6dFGS+MGS) and $z = 0.85$ (eBOSS ELG). The RIFT background model integrates the full FLRW+scalar system (equations (8)–(9)) numerically via RK45. We optimised over $(H_0, \Omega_m, r_d, \Lambda_0)$ with Nelder-Mead (20 random restarts).

Table 1: Published DM–DH correlation coefficients used in the BAO covariance matrix.

Survey	Redshift	$\rho(\text{DM}, \text{DH})$	Reference
BOSS DR12	$z = 0.38$	-0.52	Alam et al. (2017), Table 4
BOSS DR12	$z = 0.51$	-0.47	Alam et al. (2017), Table 4
eBOSS DR16 LRG	$z = 0.70$	-0.48	Alam et al. (2021), Table 3
eBOSS DR16 QSO	$z = 1.48$	-0.46	Alam et al. (2021), Table 3
Ly-α DR16	$z = 2.33$	-0.43	Bourboux et al. (2020), Table 3

Results.

$$\chi^2_{\text{diag,best}} = 11.74 \quad (\chi^2/\text{dof} = 1.47) \quad [\text{incorrect}]$$
$$\chi^2_{\text{full,best}} = \mathbf{9.78} \quad (\chi^2/\text{dof} = \mathbf{1.22}) \quad [\text{correct}]$$
$$\Delta\chi^2 = +1.95 \quad (\text{full-cov preferred})$$

Best-fit parameters: $\Omega_m = 0.294$, $\Lambda_0 = 0.003$, $H_0 \cdot r_d \approx$ const (H$_0$ and r_d individually degenerate in BAO-only fits; only $H_0 r_d$ is constrained). The $\chi^2/\text{dof} = 1.22$ result should be reported to referees; the diagonal result is statistically incorrect.

7.2 CMB Full CLASS Injection (Sim 88)

Problem with v1. The prior CMB result used a first-order modulation $C_\ell^{\text{RIFT}} \approx C_\ell^{\Lambda\text{CDM}}(1 + \epsilon)$, bypassing the full CLASS Boltzmann hierarchy and missing the acoustic-peak shift from the modified background cosmology.

Method. We derived the RIFT primordial power spectrum $P(k) = A_s(k/k_*)^{n_s-1} \cdot R^2_{\text{growth}}$ from the RIFT growth factor $D_{\text{RIFT}}(z)/D_{\Lambda\text{CDM}}(z)$, integrated via the modified linear growth equation with $G_{\text{eff}}(\Psi)$. This P(k) was injected into CLASS as `external_Pk` (format: k [Mpc^{-1}], $\Delta^2_R(k)$ dimensionless) and the full Boltzmann transfer function was computed for both RIFT (H$_0 = 68.14$, $\Omega_m = 0.294$, $\Lambda_0 = 0.003$) and ΛCDM (Planck 2018 best-fit: H$_0 = 67.36$, $\Omega_m = 0.3153$).

Results.

$$G_{\text{eff}}/G = 0.999984 \quad (16\,\text{ppm})$$
$$D_{\text{RIFT}}(z=0)/D_{\Lambda\text{CDM}}(z=0) = 1.000000 \quad (\text{to 6 d.p.})$$
$$\text{RMS}(\Delta C_\ell/C_\ell) = \mathbf{2.93}\% \quad (l = 2\text{–}1500)$$

The dominant RIFT CMB signal is the acoustic peak shift from the modified background ($H_0 = 68.14$ vs. 67.36, $\Omega_m = 0.294$ vs. 0.315), not the G_{eff} correction (16 ppm). The 2.93% RMS deviation is within Planck error bars at most multipoles.

7.3 Planck TT Likelihood (Sim 89)

Method. We implemented the approximate full-sky Gaussian TT likelihood

$$-2\ln\mathcal{L} = f_{\text{sky}}\sum_\ell (2\ell+1)\left[x_\ell - 1 - \ln x_\ell\right], \quad x_\ell = \frac{D_\ell^{\text{pred}} + N_\ell}{D_\ell^{\text{Planck}} + N_\ell}, \tag{13}$$

against the Planck 2018 best-fit TT theory spectrum (COM_PowerSpect_CMB-base-plikHM-TTTEEE R3.01), with $f_{\text{sky}} = 0.778$ (Planck plikHM TT) and Planck 143 GHz noise N_ℓ ($\Delta_T = 33\,\mu\text{K}\cdot\text{arcmin}$, $\theta_{\text{FWHM}} = 7.27'$). Note: $x_\ell = 1$ when $D_\ell^{\text{pred}} = D_\ell^{\text{Planck}}$, so $-2\ln\mathcal{L} = 0$ at the reference cosmology; the ΛCDM baseline $\chi^2 = 6.2$ validates the implementation.

Results.

Model	χ^2	Notes
ΛCDM (CLASS vs. Planck theory)	6.2	Baseline; validates formula
RIFT (CLASS vs. Planck theory)	913	BAO-only best-fit params
$\Delta\chi^2$ (RIFT$-\Lambda$CDM)	+907	$\sim 30.1\sigma$

The 30.1σ tension reflects that the RIFT BAO-only best-fit ($H_0 = 68.14$, $\Omega_m = 0.294$) predicts acoustic peaks shifted from the Planck ΛCDM spectrum. The per-mode $\chi^2 = 1.9 \times 10^{-4}$ is tiny; the large total arises from coherent accumulation over $\sim 4.9 \times 10^6$ f_{sky}-weighted modes. This is a parameter-consistency finding, not a failure of the RIFT field equations. Its resolution is the joint fit (Section 7.4).

7.4 Joint CMB+BAO Parameter Fit (Sim 90)

Method. We minimised $\chi^2_{\text{total}} = \chi^2_{\text{CMB}}(H_0, \Omega_m) + \chi^2_{\text{BAO}}(H_0, \Omega_m, r_d, \Lambda_0)$ jointly over all parameters. The CMB chi-squared uses the Gaussian approximation from the Planck 2018 published parameter covariance (Table 2 of Planck Collaboration 2020): $\chi^2_{\text{CMB}} = (\mathbf{x}-\boldsymbol{\mu})^\top C^{-1}_{\text{Planck}}(\mathbf{x}-\boldsymbol{\mu})$ with $\boldsymbol{\mu} = (67.36, 0.3153)$, $\sigma_{H_0} = 0.54$, $\sigma_{\Omega_m} = 0.0073$, $\rho = -0.90$. The BAO chi-squared uses the same 12-point data vector and 12×12 covariance matrix as Sim 87, evaluated with the flat Friedmann approximation for $H(z)$ including the first-order G_{eff} correction at representative $\Psi_0 = 0.01$ (identical method to Sim 91; the full ODE result agrees to $< 16\,\text{ppm}$ per Sim 88). LCDM: Λ_0 fixed to zero. RIFT: Λ_0 free. Nelder-Mead optimisation, 20 random restarts.

Results.

Model	H_0	Ω_m	r_d	Λ_0	$\chi^2_{\rm CMB}$	$\chi^2_{\rm BAO}$
ΛCDM	67.59	0.3118	147.56	0	0.23	11.02
RIFT	67.59	0.3118	147.57	$\approx$0.013†	0.23	11.02

† Numerically unconstrained: $\Delta\chi^2 < 0.004$ across $\Lambda_0 \in [0, 0.1]$ (Sim 91). Value depends on optimiser init.

$$\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = \mathbf{0.000} \quad \text{(PASS)}$$

The Planck CMB prior dominates the joint fit ($\sigma_{H_0} = 0.54$ km/s/Mpc is $\sim 10\times$ more constraining than BAO alone). Both RIFT and ΛCDM converge to the same optimal $(H_0, \Omega_m) \approx (67.6, 0.312)$. The coupling $\Lambda_0 = 0.003$ modifies $H(z)$ at the 16 ppm level (Sim 88), making RIFT and ΛCDM cosmologically indistinguishable at the CMB+BAO level. $\Delta\chi^2 \approx 0$ confirms that RIFT is consistent with precision cosmological data at current measurement precision, and that the Sim 89 tension was an artefact of using the BAO-only (parameter-degenerate) best-fit as the CMB input. It is important to interpret this result correctly: $\Delta\chi^2 \approx 0$ arises because at the joint best-fit coupling ($\Lambda_0 \lesssim 0.013$, $G_{\rm eff}/G = 1 - 16\,\text{ppm}$) RIFT is observationally degenerate with ΛCDM—both models converge to the same parameters and the RIFT correction is below current measurement precision. This should not be read as RIFT providing a physically distinct fit to CMB data; rather, the Sim 90 joint fit demonstrates that the theory is *consistent* with the data, not that it is *preferred* or that it explains a feature not accounted for by ΛCDM.

Note on Λ_0 reporting. Two Λ_0 values appear in the analysis: the BAO-only best-fit $\Lambda_0 = 0.003$ (Sim 87, four free parameters, physically motivated) and the joint-fit value $\Lambda_0 \approx 0.013$ (Sim 90, numerically unconstrained because the χ^2 landscape is flat in Λ_0 at this precision, $\Delta\chi^2_{\Lambda_0} < 0.004$ across $[0, 0.1]$ from Sim 91; the specific value depends on optimiser initialisation and carries no physical significance). The fiducial value quoted throughout is $\Lambda_0 = 0.003$; both are consistent with all current data.

Summary. Table 2 collects the key observational results. Best-fit parameter values for all simulations are tabulated in Appendix A.

7.5 Non-linear Structure Growth and σ_8 (Sim 92)

Method. We computed the linear and non-linear matter power spectra for the SIM90 joint best-fit cosmology ($H_0 = 67.59$ km/s/Mpc, $\Omega_m = 0.312$, $\Omega_b = 0.049$, $n_s = 0.9649$, $A_s = 2.101 \times 10^{-9}$) across 15 values of $\Lambda_0 \in [0, 0.1]$. The linear $P(k)$ was generated by CLASS at $z = 0$ with the RIFT modification $P_{\rm RIFT}(k) = P_{\Lambda\rm CDM}(k) \times (D_{\rm RIFT}/D_{\Lambda\rm CDM})^2$, where the growth factor ratio is taken from Sim 88. Non-linear corrections were applied via the Takahashi et al. (2012) HALOFIT fitting formula [Takahashi et al., 2012]. The matter fluctuation amplitude is

$$\sigma_8^2 = \int_0^\infty \frac{k^2\,dk}{2\pi^2} P_{\rm lin}(k) \left[\frac{3j_1(k \cdot 8\,h^{-1}\text{Mpc})}{k \cdot 8\,h^{-1}\text{Mpc}}\right]^2, \tag{14}$$

computed on $k \in [10^{-4}, 10^2]\,h\,\text{Mpc}^{-1}$ with 2000 log-spaced points. The weak-lensing amplitude $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ is evaluated at the joint best-fit $\Omega_m = 0.312$.

Results. At the LCDM baseline ($\Lambda_0 = 0$), the joint best-fit cosmology gives $\sigma_8^{\text{lin}} = 0.824$, consistent with the Planck 2018 constraint $\sigma_8 = 0.811 \pm 0.006$ [Planck Collaboration, 2020] at the 2.2σ level (expected given our simplified CMB Gaussian prior). The RIFT coupling suppresses the linear growth factor, reducing σ_8 proportionally to $D^2_{\text{RIFT}}/D^2_{\Lambda\text{CDM}}$:

Λ_0	$D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$	σ_8^{lin}	$\Delta\sigma_8$	S_8
0.000	1.000000	0.8239	0.000%	0.840
0.003	0.999931	0.8238	−0.007%	0.840
0.008	0.999815	0.8237	−0.018%	0.840
0.050	0.998845	0.8229	−0.116%	0.839
0.100	0.997691	0.8220	−0.231%	0.838

The Planck/KiDS-1000 σ_8 tension amounts to ~ 5–6% in σ_8 (KiDS-1000: $S_8 = 0.766 \pm 0.020$, Asgari et al. 2021; DES Y3: $S_8 = 0.776 \pm 0.017$, Abbott et al. 2022). The RIFT suppression at the BAO best-fit coupling ($\Lambda_0 = 0.003$) is $\Delta\sigma_8 = -0.007\%$— two orders of magnitude too small to alleviate this tension. Even at the detection threshold ($\Lambda_0 = 0.05$), $\Delta\sigma_8 = -0.116\%$, still $50\times$ below the observed discrepancy.

Conclusion. RIFT modifies σ_8 only through the linear growth factor. At physically motivated coupling strengths ($\Lambda_0 \leq 0.05$), the modification is $|\Delta\sigma_8| < 0.12\%$, and the Planck/KiDS σ_8 tension is *not alleviated*. This is a falsifiable prediction: RIFT makes no claim to resolve the σ_8 tension at current observational precision.

7.6 DESI Year 1 BAO Refit (Sim 94)

Motivation. Sim 87 established RIFT consistency with the BOSS+eBOSS BAO dataset ($\chi^2/\text{dof} = 1.22$). The DESI Collaboration has since released Year 1 BAO measurements [Adame et al., 2024] spanning 13 data points across 7 tracer bins ($0.3 \leq z_{\text{eff}} \leq 2.33$), providing an independent cross-check on the RIFT fit and the joint CMB+BAO parameter solution.

Data. We use the DESI Y1 BAO measurements from Table 1 of Adame et al. [2024]: BGS ($z_{\text{eff}} = 0.295$, D_V/r_d); LRG ($z = 0.51, 0.71$); LRG+ELG ($z = 0.93$); ELG ($z = 1.32$); QSO ($z = 1.49$); Lya QSO ($z = 2.33$), with the published within-bin DM–DH correlation coefficients $\rho \in [-0.49, -0.39]$. The covariance is block-diagonal (one 2×2 block per paired bin, zero between bins), following the published survey geometry.

Results.

1. **BAO-only fit:** LCDM achieves $\chi^2/\text{dof} = 13.6/10 = 1.36$ on the DESI data; RIFT gives an identical χ^2 ($\Delta\chi^2 = 0$), consistent with the finding (Sim 91) that Λ_0 does not enter $H(z)$ at detectable levels for $\Lambda_0 \leq 0.1$. The slightly elevated χ^2/dof relative to BOSS (1.22) is driven by known 2–3σ tensions in the DESI LRG bins ($z = 0.51$ D_H/r_d: -2.5σ; $z = 0.71$ D_M/r_d: -2.0σ), a feature of the DESI data independent of the gravity model.

2. **Joint DESI+CMB fit:** Adding the Planck 2018 Gaussian prior, the joint minimum gives $H_0 = 67.66$ km/s/Mpc, $\Omega_m = 0.311$, $r_d = 148.5$ Mpc, $\Lambda_0 = 0.063$

(unconstrained); $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) \approx 0$. These parameters are consistent with the SIM90 BOSS-based joint fit ($H_0 = 67.59$, $\Omega_m = 0.312$) at the $\lesssim 0.2\sigma$ level.

3. Λ_0 **sensitivity:** The DESI BAO χ^2 is flat across $\Lambda_0 \in [0, 0.1]$ ($\Delta\chi^2_{\text{max}} < 10^{-4}$), confirming that the RIFT field modification is undetectable at current BAO precision regardless of the dataset used.

Conclusion. RIFT passes the DESI Y1 BAO test with $\Delta\chi^2 = 0$ relative to ΛCDM. The joint CMB+DESI parameter solution is fully consistent with the BOSS-based SIM90 result, confirming the stability of RIFT parameters across independent BAO datasets. Λ_0 remains a null parameter at current observational precision.

7.7 CMB Polarization EE/TE (Sim 95)

Motivation. Simulations 88–89 established RIFT consistency with the Planck 2018 temperature (TT) power spectrum. Sim 95 extends this test to E-mode polarization: the EE auto-power spectrum and the TE cross-power spectrum. Polarization is independently generated at recombination and provides a complementary probe of the acoustic-peak structure, reionization optical depth, and matter content.

Method. We run the CLASS Boltzmann code with `output = tCl,pCl,lCl` for three cosmologies: (A) the RIFT BAO-only best-fit (SIM87: $H_0 = 68.14$, $\Omega_m = 0.294$), (B) the RIFT joint CMB+BAO best-fit (SIM90: $H_0 = 67.59$, $\Omega_m = 0.312$), and (C) the LCDM Planck 2018 best-fit baseline. The RIFT P(k) is constructed from the growth-factor ratio $R = D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$ (identical to Sim 88). The lensed C_ℓ^{EE} and C_ℓ^{TE} are compared against the Planck 2018 published best-fit theory spectrum (`COM_PowerSpect_CMB-base-plikHM-TTTEEE R3.01`) over $\ell = 2$–1996.

Results.

1. **Growth factor:** $D_{\text{RIFT}}(z = 0)/D_{\Lambda\text{CDM}}(z = 0) = 1.000000$ at both coupling values ($\Lambda_0 = 0.003$ and 0.013). The G_{eff} correction ($\lesssim 65$ ppm) is negligible in all polarization spectra.

2. **Joint best-fit RMS deviations** (RIFT vs LCDM Planck, $\ell = 2$–1996): $\text{RMS}(\Delta C_\ell^{EE}/C_\ell^{EE}) = 0.19\%$; $\text{RMS}(\Delta C_\ell^{TE}/\text{RMS}(C_\ell^{TE})) = 0.43\%$; $\text{RMS}(\Delta C_\ell^{TT}/C_\ell^{TT}) = 0.20\%$. All are well below Planck's measurement uncertainties at most multipoles. The small deviations arise from the (H_0, Ω_m) offset between the RIFT joint best-fit and the Planck LCDM reference point.

3. **Approximate Planck polarization likelihood:** Using a Gaussian approximation over 1995 polarization modes, the LCDM baseline gives $\chi^2_{\text{EE}} = 57.6$, $\chi^2_{\text{TE}} = 37.2$; the RIFT joint best-fit gives $\chi^2_{\text{EE}} = 89.0$, $\chi^2_{\text{TE}} = 60.0$ ($\Delta\chi^2_{\text{EE}} = +31.5$, $\Delta\chi^2_{\text{TE}} = +22.8$). These excesses arise from the same parameter offset that generates the small TT deviation after the joint fit.

4. **BAO-only tension in polarization:** The RIFT BAO-only best-fit produces $\chi^2_{\text{EE}} = 4353$ and $\chi^2_{\text{TE}} = 3110$ vs Planck, confirming that the TT tension identified in Sim 89 ($\Delta\chi^2_{\text{TT}} = 907$) propagates to all CMB spectra equally. This is consistent with the tension being driven by the (H_0, Ω_m) parameter-space shift, not by the RIFT field equations.

Conclusion. RIFT at the joint CMB+BAO best-fit reproduces Planck EE and TE with sub-percent fractional accuracy ($< 0.2\%$ for EE, $< 0.5\%$ for TE). The polarization results confirm: (i) $G_{\rm eff}$ modifications are invisible in polarization; (ii) the BAO-CMB tension is a parameter-space effect present in all spectra; (iii) the joint-fit parameter solution restores consistency across TT, EE, and TE simultaneously.

7.8 RSD Growth Rate $f\sigma_8$ (Sim 96)

Redshift-space distortions (RSD) measure the linear growth rate $f(z) = d\ln D/d\ln a$ through the combination $f\sigma_8(z)$, where $D(z)$ is the linear growth factor normalized to unity at $z = 0$ and $\sigma_{8,0} = 0.824$ is taken from the SIM90 joint best-fit. In RIFT the gravitational coupling governing growth is modified to $G_{\rm eff} = G/(1 + 16\pi\Lambda(\Psi))$, entering the growth equation as

$$D'' + \left(2 + \frac{d\ln H}{d\ln a}\right) D' = \frac{3}{2}\frac{\Omega_m}{a^3 E^2}\, G_{\rm eff}\, D, \tag{15}$$

where primes denote $d/d\ln a$ and $E(z) = H(z)/H_0$.

Data. We use the gold-sample RSD compilation spanning $z = 0.067$–1.48 (Table 3): 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 ($z = 0.38, 0.51, 0.61$) [Alam et al., 2017], VIPERS [Pezzotta et al., 2017], eBOSS DR16 LRG [Gil-Marín et al., 2020], eBOSS DR16 ELG [de Mattia et al., 2021], and eBOSS DR16 QSO [Neveux et al., 2020]. All model predictions use the SIM90 cosmological background with no additional free parameters.

Results. At the SIM90 joint best-fit ($H_0 = 67.59$, $\Omega_m = 0.312$, $\Lambda_0 \approx 0.013$) RIFT yields $\chi^2/\text{dof} = 7.73/9 = 0.86$, compared to $7.87/9 = 0.87$ for ΛCDM Planck 2018, giving

$$\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = -0.14 \approx 0. \tag{16}$$

All individual pulls are within 2σ. The maximum fractional deviation of the RIFT $f\sigma_8$ curve from the ΛCDM curve at the best-fit is 0.62% (across $z = 0$–2), which is sub-percent and well below current RSD measurement precision.

$G_{\rm eff}$ suppression. At $\Lambda_0 \approx 0.013$ the correction $16\pi\Lambda(\Psi) = 16\pi \times 0.013 \times (0.01)^2 \approx 6.5 \times 10^{-5}$, so $G_{\rm eff}$ deviates from G by ≈ 65 ppm. The observed $\sim$0.6% difference in $f\sigma_8$ between RIFT joint and ΛCDM arises entirely from the (H_0, Ω_m) parameter offset, not from the recursive field.

Λ_0 scan. A scan of $\Lambda_0 \in [0, 0.2]$ (holding all other parameters at SIM90 joint values) shows the $f\sigma_8$ deviation from ΛCDM remains below 1% for $\Lambda_0 < 0.2$. The detection threshold exceeds the Bayesian upper limit $\Lambda_0 < 0.095$ (95% CL, Sim 93), so current RSD data cannot independently constrain Λ_0.

S_8 tension context. With $\sigma_{8,0} = 0.824$ and $\Omega_m = 0.312$, the RIFT prediction $S_8 = \sigma_8\sqrt{\Omega_m/0.3} = 0.840$ lies 3.7σ above KiDS-1000 ($S_8 = 0.766 \pm 0.020$, Asgari et al. 2021), compared to 3.9σ for ΛCDM Planck. RIFT provides no relief for the S_8 tension, consistent with the SIM92 result ($\Delta\sigma_8 < 0.007\%$ at $\Lambda_0 = 0.003$).

Conclusion. RIFT passes all current RSD / $f\sigma_8$ constraints. The growth rate is indistinguishable from ΛCDM at the joint best-fit because $G_{\text{eff}} \approx G$ to better than 10 ppm. Detection of the RIFT growth signature requires $\Lambda_0 > 0.2$, which is excluded by the BAO+CMB joint fit, making the growth rate a consistency test rather than a discriminator with current data. Future surveys (DESI Y5, Euclid) measuring $f\sigma_8$ to $\sim$1% at $z > 1$ will tighten this constraint to $\Lambda_0 \lesssim 0.05$.

7.9 Definitive Planck 2018 plikHM TTTEEE Likelihood (Sim 97)

Motivation. Simulations 89 and 95 compared RIFT Boltzmann spectra against the Planck published theory curve using an approximate Gaussian CMB likelihood. Sim 97 replaces this with the *official* Planck 2018 plikHM TTTEEE likelihood code, accessed via the `clipy` v0.15 Python interface to the `clik` library [Planck Collaboration, 2020]. This is the identical likelihood used in the Planck 2018 cosmological parameter analysis.

Method. Three cosmologies are evaluated against the full official Planck 2018 likelihood: (A) RIFT joint CMB+BAO best-fit (Sim 90: $H_0 = 67.59$, $\Omega_m = 0.312$, $\Lambda_0 = 0.008$, original value at time of Sim 97 run); (B) RIFT BAO-only best-fit (Sim 87: $H_0 = 68.14$, $\Omega_m = 0.294$, $\Lambda_0 = 0.003$); (C) ΛCDM Planck 2018 best-fit [Planck Collaboration, 2020].

For each cosmology, CLASS v3 computes the lensed TT, EE, and TE spectra with the RIFT background (same pipeline as Sim 95). The C_ℓ vectors are evaluated by:

- **plikHM TTTEEE**: high-ℓ ($\ell = 30$–2508) likelihood with 47 nuisance parameters (foreground amplitudes, calibration) fixed at Planck best-fit values from the `clik` self-check vector;
- **lowl TT (Commander)**: low-ℓ ($\ell = 2$–29) TT;
- **lowl EE (SimAll)**: low-ℓ ($\ell = 2$–29) EE polarization.

Pipeline validation. ΛCDM gives $-2\ln\mathcal{L}_{\text{plikHM}} = 2352.05$, compared to the `clik` self-check value $2 \times 1172.47 = 2344.94$ (agreement to $< 0.3\%$, attributable to our parameter values not being bit-for-bit identical to the self-check cosmology).

Results.

Cosmology	plikHM TTTEEE	lowl TT	lowl EE	Total $-2\ln\mathcal{L}$
RIFT joint (A)	2359.05	23.54	396.05	2778.63
RIFT BAO-only (B)	3297.80	24.26	395.93	3717.99
ΛCDM (C)	2352.05	23.42	396.14	2771.61

$$\Delta(-2\ln\mathcal{L})_{\text{plikHM}}(\text{RIFT joint} - \Lambda\text{CDM}) = +7.0 \tag{17}$$
$$\Delta(-2\ln\mathcal{L})_{\text{plikHM}}(\text{RIFT BAO-only} - \Lambda\text{CDM}) = +945.7 \tag{18}$$

Interpretation. The BAO-only tension ($\Delta(-2\ln\mathcal{L}) = +946$, Eq. 18) is fully consistent with the Sim 89 approximate result ($\Delta\chi^2 = +907$), cross-validating both pipelines.

The RIFT joint best-fit (A) shows a $\Delta(-2\ln\mathcal{L}) = +7.0$ excess (Eq. 17). This arises from the (H_0, Ω_m) offset between the Sim 90 joint parameters (optimised against the approximate Gaussian CMB prior) and the true Planck CMB optimum. At $\Lambda_0 = 0.008$, $G_{\rm eff}/G = 1 - 42\,\text{ppm}$ contributes negligibly to any CMB observable; the +7.0 excess is entirely parameter-driven. Specifically, RIFT evaluated at the Planck ΛCDM parameter point ($H_0 = 67.36$, $\Omega_m = 0.3153$, $\Lambda_0 = 0.008$) yields $\Delta(-2\ln\mathcal{L}) \approx 0$, since $G_{\rm eff}$ deviates from G by only 42 ppm.

In frequentist terms, $\Delta(-2\ln\mathcal{L}) = +7.0$ with one effective degree of freedom (Λ_0 being the extra parameter) corresponds to $\approx 2.6\sigma$ tension—a marginal result. This tension is resolved by Sim 98 below, which re-optimises the RIFT parameter fit using the real plikHM likelihood.

Conclusion. The RIFT *theory* is fully consistent with the real Planck 2018 plikHM TTTEEE likelihood: the $G_{\rm eff}$ correction at the joint best-fit coupling is 42 ppm and invisible to the CMB. The marginal $\Delta(-2\ln\mathcal{L}) = +7.0$ is a parameter-optimisation artefact that vanishes in Sim 98.

7.10 Joint plikHM + BAO Parameter Re-fit (Sim 98)

Motivation. Sim 97 revealed a $\Delta(-2\ln\mathcal{L}) = +7.0$ excess at the Sim 90 joint best-fit parameters. The cause was identified as the (H_0, Ω_m) offset between Sim 90 (optimised against an approximate Gaussian CMB prior) and the true Planck CMB optimum. Sim 98 resolves this by re-running the joint CMB+BAO optimisation using the real plikHM TTTEEE likelihood.

Method. We minimise

$$\chi^2_{\rm total}(H_0, \Omega_m[, \Lambda_0]) = -2\ln\mathcal{L}_{\rm plikHM} + (-2\ln\mathcal{L}_{\rm lowl\,TT}) + (-2\ln\mathcal{L}_{\rm lowl\,EE}) + \chi^2_{\rm BAO} \tag{19}$$

over (H_0, Ω_m) for ΛCDM and $(H_0, \Omega_m, \Lambda_0)$ for RIFT, using Nelder-Mead optimisation (max 80 CLASS evaluations each). BAO data: BOSS DR12 + eBOSS DR16 + Lyα with full 12×12 covariance (same as Sim 87). The sound horizon r_d is computed analytically from $(H_0, \Omega_m, \Omega_b)$ via numerical integration of $c_s(a)/[a^2 H(a)]$ to the baryon drag epoch (Eisenstein & Hu 1998 approximation for $z_{\rm drag}$).

Results.

Model	H_0	Ω_m	Λ_0	r_d	plikHM $-2\ln\mathcal{L}$	$G_{\rm eff}/G$
ΛCDM	67.30	0.3169	0	150.6	2347.46	1
RIFT	67.30	0.3168	0.00506	150.6	2347.45	$1 - 24\,\text{ppm}$

$$\Delta(-2\ln\mathcal{L})_{\rm plikHM}(\text{RIFT} - \Lambda\text{CDM}) = -0.013 \approx 0 \quad (\textbf{PASS}) \tag{20}$$

The RIFT and ΛCDM best-fits are essentially identical: $\Delta H_0 = +0.002$, $\Delta\Omega_m = -0.0001$, $\Delta\Lambda_0 = 0.005$ (with $G_{\rm eff}$ deviation 24 ppm). The $\chi^2_{\rm total}$ difference is +0.004.

Progression of CMB+BAO tests.

Simulation	$\Delta(-2\ln\mathcal{L})_{\text{plikHM}}$	Parameter point	Notes
Sim 89 (approx Gaussian)	$+907$	BAO-only best-fit	Parameter tension
Sim 90 (approx Gaussian)	≈ 0	Joint approx best-fit	Resolved by joint fit
Sim 97 (real plikHM)	$+7.0$	Sim 90 approx params	Residual offset
Sim 98 (real plikHM)	$\mathbf{-0.013}$	Real plikHM best-fit	**Fully resolved**

Conclusion. When RIFT parameters are optimised against the real Planck 2018 plikHM TTTEEE likelihood, $\Delta(-2\ln\mathcal{L}) = -0.013 \approx 0$. The Sim 97 tension of $+7.0$ is completely resolved. RIFT is fully consistent with the official Planck 2018 CMB likelihood at the optimised parameter point, with $G_{\text{eff}}/G = 1 - 24\,\text{ppm}$.

7.11 Galaxy Rotation Curves: Honest Test (Sim 99)

Sim 99 is the first quantitative test of the RIFT dark-matter prediction using the locked action. Two well-observed flat-curve spirals are used: NGC 3198 ($v_{\text{flat}} = 157$ km/s; de Blok et al. 2008) and NGC 6503 ($v_{\text{flat}} = 116$ km/s; Begeman 1987).

Setup. The static, spherically-symmetric field equation derived from the locked action (eq. 1) is

$$\Psi'' + \frac{2}{r}\Psi' = m_0^2\Psi + \frac{\lambda_{\text{gal}}}{1}\Psi^3 + 2\Lambda_0\Psi\, R(r), \tag{21}$$

where $R(r) \approx -8\pi(G/c^2)\rho_{\text{bar}}(r)$ is the weak-field Ricci scalar sourced by the observed baryonic distribution, and $V(\Psi) = (\lambda_{\text{gal}}/4)\Psi^4$ is the simplest nonlinear extension consistent with $V'(0) = 0$ (no constant background drive) and renormalizability. The effective dark-matter density is $\rho_\Psi = (c^2/G)\big[\frac{1}{2}(\Psi')^2 + \frac{1}{2}m_0^2\Psi^2 + (\lambda_{\text{gal}}/4)\Psi^4\big]$.

The physical boundary condition selects the *decaying* Yukawa halo mode $\Psi \sim A\,e^{-m_0 r}/r$ by integrating inward from $r_{\max}$:

$$\Psi(r_{\max}) = \Psi_{\text{bc}}, \qquad \Psi'(r_{\max}) = -\left(m_0 + \frac{1}{r_{\max}}\right)\Psi_{\text{bc}}. \tag{22}$$

This shooting method avoids the growing-mode contamination ($\Psi \sim e^{+m_0 r}/r$) that produced unphysical velocities in preliminary runs.

Scan. Sim 99 scans 30 combinations of $(\Psi_{\text{bc}}, \lambda_{\text{gal}})$: $\Psi_{\text{bc}} \in \{10^{-8}, 10^{-6}, 10^{-4}, 5\times10^{-4}, 10^{-3}\}$ and $\lambda_{\text{gal}} \in \{-10, -1, 0, +1, +10, +100\}$, at the cosmological best-fit $\Lambda_0 = 0.003$. A solution is accepted only if: (i) the rotation curve satisfies $|v_c(r) - v_{\text{flat}}|/v_{\text{flat}} < 20\%$ for $r > r_{\max}/2$; (ii) the cosmological bound $\Lambda_0\Psi^2 < \Lambda_0^{\text{bound}}/(16\pi)$ is satisfied everywhere; and (iii) $\rho_\Psi \geq 0$ everywhere.

Result: FAIL. No combination passes. The physical reason is geometric: the decaying Yukawa profile at $m_0 = 0.1\,\text{kpc}^{-1}$ gives $\rho_\Psi(r) \propto e^{2m_0(r_{\max}-r)}/r^2$, which rises $\sim$ 150-fold from $r = 30$ kpc to $r = 5$ kpc—exponentially steeper than the $1/r^2$ isothermal profile required for a flat rotation curve. The quartic interaction $V = (\lambda_{\text{gal}}/4)\Psi^4$ modifies the central amplitude but cannot alter this exponential radial dependence. At $\Lambda_0 = 0.003$

(cosmological best-fit), G_{eff}/G deviates from unity by < 10 ppm—the G_{eff} channel likewise provides zero flattening.

Interpretation. The locked RIFT action at cosmological parameters does not replace dark matter via T^{Ψ} or G_{eff}. Galactic-scale predictions require either a screening mechanism that suppresses the Yukawa mass m_0 on galactic scales, or a condensation mechanism that produces a self-sustained $\rho \propto 1/r^2$ profile. Neither is present in the current locked action. Rotation curves therefore remain a falsifiable future prediction, not an achieved result.

7.12 Sim 100: Time-Domain Confirmation of the Static Limit

Sim 99 solved the *static* Yukawa equation. Sim 100 removes this restriction: it evolves the full 1+1D spherical wave equation

$$\frac{\partial^2\Psi}{\partial t^2} = \frac{\partial^2\Psi}{\partial r^2} + \frac{2}{r}\frac{\partial\Psi}{\partial r} - m_0^2\Psi - \lambda\Psi^3 + 2\Lambda_0\Psi\, R(r,t) \tag{23}$$

via method-of-lines with an implicit Radau solver ($N_r = 120$, $\Delta r = 0.25$ kpc). Initial condition: the uniform cosmological background $\Psi(r,0) = \Psi_{\text{cosmo}} = 0.003$ at rest ($\partial_t\Psi = 0$). Boundary conditions: Neumann at $r_{\min}$ (spherical symmetry), Dirichlet $\Psi(r_{\max}) = \Psi_{\text{cosmo}}$ at $r_{\max} = 30$ kpc. Integration runs to $t_{\text{end}} = 150\,\text{kpc}/c \approx 489$ kyr ≈ 2.4 Compton periods.

Curvature coupling confirmed negligible. The galactic Ricci scalar peaks at $|R_{\text{gal}}| \approx 8.5\times10^{-7}\,\text{kpc}^{-2}$ (at $r = 0.5$ kpc), giving a coupling $2\Lambda_0|R_{\text{gal}}| = 5.1\times10^{-9}\,\text{kpc}^{-2} \ll m_0^2 = 0.01\,\text{kpc}^{-2}$. Even with a proto-galactic overdensity $\delta = 100$, this coupling is 5×10^7 times smaller than m_0^2. Condensation driven by the $2\Lambda_0\Psi R$ term is impossible at galactic densities; neutron-star densities ($\rho \sim 10^{15}\,M_\odot\,\text{kpc}^{-3}$) would be required.

Attractive self-coupling scan. To test whether the quartic interaction can drive condensation, we scan $\lambda \in \{0, -100, -200, -400, -600, -800, -1000, -1500\}$ and $\delta_{\max} \in \{0, 100\}$ (16 total runs). The tachyonic instability threshold for the lowest spatial mode is

$$\lambda_{\text{crit}}^{\text{spatial}} = -\frac{(k_{\min}^2 + m_0^2)}{3\Psi_{\text{cosmo}}^2} \approx -778, \qquad k_{\min} = \frac{\pi}{r_{\max}}, \tag{24}$$

and the global blow-up threshold for the uniform mode (potential barrier) is $\lambda_{\text{blow}} = -m_0^2/\Psi_{\text{cosmo}}^2 \approx -1111$.

Results: for $\lambda \in [0, -1000]$ the field remains stable (growth factor $\leq 1.21\times$), relaxing from the uniform IC toward the static Yukawa profile of Sim 99—confirming the adiabatic approximation (Compton time 33 kyr $\ll t_{\text{form}} \sim 1$ Gyr). For $\lambda = -1500$ (beyond the global blow-up threshold), the field grows without bound within $< 10\,\text{kpc}/c$: the potential $V = (\lambda/4)\Psi^4$ is unbounded below and has no stable condensate minimum. In no run does $v_c(r)$ come within 20% of v_{flat}.

Conclusion. The time-domain simulation confirms Sim 99 with a direct structural argument: $V = (\lambda/4)\Psi^4$ with $\lambda < 0$ is an inverted quartic with no lower bound. It either leaves the field oscillating around $\Psi = 0$ (sub-threshold, $|\lambda| < 1111$, Yukawa profile unchanged) or causes runaway blow-up (super-threshold, $|\lambda| > 1111$). There is no window of λ that produces a stable self-bound halo with a flat rotation curve. A physically stable condensation channel would require a sextic or higher stabilizing term not present in the current locked action.

8 Recursive Graviton Emission

In RIFT, gravitational waves are emitted not from accelerated masses, but from curvature memory spikes that surpass feedback thresholds. These events generate quantized ripple packets: *recursive gravitons.*

Recursive graviton emission occurs when the memory tensor spike exceeds the threshold

$$\delta\mathcal{M}_{\mu\nu} \geq \frac{1}{\eta_c}\frac{d}{d\tau}\left(\nabla_{(\mu}\phi_{\nu)}\right), \tag{25}$$

where ϕ_μ is the recursive vector potential from Ψ, η_c is the memory coupling constant, and τ is proper time.

Each recursive graviton carries energy $E = \hbar\omega\int|\delta\mathcal{M}_{\mu\nu}|^2\,d^3x$.

Physical features. Emission is discrete (triggered when curvature memory spikes), spin-2 symmetry is preserved, propagation slows in memory-rich zones due to decoherence, and the source is geometric rather than energetic.

Testable predictions. Echo-phase ripples in void regions, B-mode polarization with recursive ring harmonics, lensing interference from memory collapse (not binary mergers), nonthermal gravitational bursts unaligned with mass-based models.

9 Falsifiability Predictions

RIFT stands apart from speculative models by offering clear, testable predictions with well-defined thresholds for falsification.

9.1 Galaxy Rotation Curves

Prediction: Flat curves arise from recursive curvature—not dark matter. *Sim 99–100 status:* The locked action at $\Lambda_0 = 0.003$ *does not* produce flat rotation curves via T^Ψ or G_{eff} in either the static (Sim 99) or time-domain (Sim 100) treatment. The decaying Yukawa profile ($m_0 = 0.1\,\text{kpc}^{-1}$) is exponentially steeper inward than the $1/r^2$ isothermal profile required for curve flattening (Section 7.11). The galactic Ricci coupling $2\Lambda_0|R_{\text{gal}}| \approx 5\times10^{-9}\,\text{kpc}^{-2}$ is 5×10^7 times weaker than m_0^2: curvature-driven condensation is ruled out at galactic densities. Attractive self-coupling ($\lambda < 0$) either leaves the Yukawa profile unchanged (sub-threshold) or causes tachyonic blow-up (super-threshold), with no stable halo condensate in between (Section 7.12). The DM-replacement prediction therefore requires derivation of a galactic-scale screening or higher-order stabilization mechanism not yet present in the locked action. *Test:* Reconstruct $\Psi(r)$ from observed curves once such a mechanism is derived; verify that field gradient sustains rotation. *Falsification:* If no such mechanism can be derived from the locked action, the DM-replacement prediction is falsified.

9.2 Void Lensing (DES, Euclid)

Prediction: Weak lensing without mass near voids, via phase-delayed geodesics from residual memory $\mathcal{M}_{\mu\nu}$. *Test:* Cross-correlate Planck lensing with DES voids; look for phase shifts. *Falsification:* No lensing where RIFT predicts curvature $\Rightarrow$ falsification.

9.3 Bullet Cluster Lensing Offset

Prediction: Lensing offset explained by curvature memory persisting after mass displacement. *Test:* Simulate lensing profiles with recursive memory; compare to κ maps. *Falsification:* Cannot reproduce lensing peak without dark mass.

9.4 CMB Low-ℓ Anomalies

Prediction: Recursive memory decay suppresses low-ℓ power. *Test:* Compare RIFT C_ℓ to Planck 2018. Validate suppression shape. *Falsification:* Cannot match low-ℓ drop without fine-tuning.

10 BAO Sound Horizon from First Principles (Sim 101)

Motivation. Section 6.2 described BAO in RIFT via recursive echo shells but noted that the sound horizon r_d was a free parameter fitted to data, not derived from the field equations. Sim 101 closes this gap by integrating the full RIFT scalar-field system through recombination and computing r_d from first principles.

Method. The comoving sound horizon is

$$r_d = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)\, c}{H(z)}\, dz\,, \tag{26}$$

where $c_s^2 = c^2/3\,(1+R_b)^{-1}$, $R_b = 3\rho_b/(4\rho_\gamma)$, and $z_{\rm drag} = 1059$ (CLASS-calibrated for the SIM90 cosmology). RIFT modifies $H(z)$ through the scalar stress-energy:

$$H^2_{\rm RIFT}(z) = H^2_{\Lambda{\rm CDM}}(z)\left(1 + \frac{\rho_\Psi}{\rho_{\rm crit}}\right), \qquad \rho_\Psi = \tfrac{1}{2}\dot{\Psi}^2 + \tfrac{1}{2}m_0^2\Psi^2\,, \tag{27}$$

and c_s via $G_{\rm eff} = G/(1+16\pi G\Lambda_0\Psi^2)$. The scalar $\Psi(z)$ is obtained by integrating the Klein-Gordon equation in the z-variable from $z_{\rm max} = 10^6$ to $z = 0$:

$$\frac{du}{dz} = \left[\frac{2}{1+z} - \frac{H'(z)}{H(z)}\right] u - \frac{m_0^2\,\Psi}{(1+z)^2 H^2(z)}\,, \qquad u \equiv \frac{d\Psi}{dz}\,, \tag{28}$$

with initial conditions $\Psi(z_{\rm max}) = \Psi_0 = 0.003$, $u(z_{\rm max}) = 0$ (field frozen at early times when $m_0 \ll H(z_{\rm max})$). The ODE is integrated with the Radau solver at rtol $= 10^{-9}$.

Results.

Λ_0	$r_d^{\rm RIFT}$ [Mpc]	Δr_d [Mpc]	$\Delta r_d/r_d$	$\rho_\Psi/\rho_{\rm tot}$ at $z_{\rm drag}$
0 (LCDM)	153.05	0	0	—
0.003	153.05	-0.000104	-0.68 ppm	1.5×10^{-10}
0.008	153.05	-0.000277	-1.8 ppm	1.5×10^{-10}
0.050	153.05	-0.001731	-11 ppm	1.5×10^{-10}
0.095	153.05	-0.003289	-21 ppm	1.5×10^{-10}

The LCDM baseline $r_d = 153$ Mpc differs from the Planck value of 147 Mpc because our simplified Friedmann equation omits neutrino free-streaming ($\Delta N_{\rm eff}$); both RIFT and LCDM share this approximation, so the fractional shift is unaffected.

The maximum shift across the entire allowed range $\Lambda_0 \in [0, 0.095]$ is $|\Delta r_d/r_d| =$ 21 ppm, compared to current BAO measurement precision of ~ 3000 ppm (0.3%). RIFT and LCDM are degenerate in r_d at all observationally allowed couplings.

Interpretation. The negligible $\rho_\Psi/\rho_{\rm tot} \sim 10^{-10}$ at the drag epoch confirms that the scalar field carries essentially no energy density during recombination: the RIFT background is indistinguishable from the FLRW background at $\Lambda_0 \ll 1$. The echo-shell picture of Section 6.2 is therefore a *physical interpretation* of standard photon-baryon acoustic physics, not a first-principles derivation of a distinct r_d. This closes the gap identified by reviewer P2.7: the RIFT BAO mechanism does not predict a different sound horizon from ΛCDM, and r_d remains a free parameter constrained by data. The detection threshold for a RIFT r_d signal requires $\Lambda_0 \gg 0.1$, well beyond the current 95% upper limit.

11 One-Loop UV Structure (Sim 102)

Motivation. Section 5 conjectured that the memory-tensor commutator decay $[\mathcal{M}_{\mu\nu}(x), \mathcal{M}_{\rho\sigma}(y)] \propto e^{-|t_x - t_y|/\tau_m}$ renders loop diagrams UV-finite, but no explicit calculation was performed. Sim 102 carries out that calculation for the leading one-loop contribution to the Ψ self-energy from the non-minimal coupling $\Lambda_0 \Psi R$.

Setup. In linearized gravity ($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$), the Ricci scalar at linear order in h is $R(k) \sim k^2 \tilde{h}(k)$ in Fourier space (harmonic gauge). The $\Lambda_0 \Psi R$ vertex therefore contributes a factor $V(k) \sim \Lambda_0 k^2$ to the Feynman rules. The graviton propagator in the same gauge is $D_{\rm grav}(k) = i/k^2$. The memory damping enters as a form factor on the graviton propagator:

$$D_{\rm RIFT}(k) = \frac{e^{-2k^2/k_m^2}}{k^2}\,, \tag{29}$$

where $k_m = 1/\tau_m$ is the memory wavenumber cutoff.

The one-loop Ψ self-energy in Euclidean space (after Wick rotation) reduces, by $O(4)$ symmetry, to a one-dimensional radial integral:

$$\Sigma(0) = \frac{\Lambda_0^2}{8\pi^2} \int_0^\infty dk\, \frac{k^5\, e^{-2k^2/k_m^2}}{k^2 + m_0^2}\,. \tag{30}$$

Results: UV divergence without memory. Setting $e^{-2k^2/k_m^2} \to 1$ (no memory, $k_m \to \infty$) and imposing a hard cutoff $k_{\rm UV}$:

$$\Sigma_{\rm bare}(0) \propto \Lambda_0^2\, k_{\rm UV}^4\,, \tag{31}$$

confirmed numerically with power-law slope 4.000 ± 0.001 across $k_{\rm UV} \in [1, 1000]\,\text{Mpc}^{-1}$. This is a quartic UV divergence, as expected from power counting ($\sim d^4k \cdot k^2/k^2 \to k^4$).

Results: finiteness with memory. With the damping factor in eq. (29), the integral (30) converges for any finite k_m. In the limit $k_m \gg m_0$:

$$\Sigma(0) \;\approx\; \frac{\Lambda_0^2\, k_m^4}{64\pi^2}\,, \tag{32}$$

verified numerically to $< 0.22\%$ accuracy for $k_m > 10\, m_0$. The effective radiative mass shift is therefore $\delta m^2 = \Sigma(0) \sim \Lambda_0^2 k_m^4/(64\pi^2)$.

k_m [Mpc^{-1}]	$\Sigma_{\rm num}(0)$ [Mpc^{-2}]	$\Sigma_{\rm analytic}$	ratio	$\delta m^2/m_0^2$
0.01	6.3×10^{-17}	1.4×10^{-16}	0.45	6.3×10^{-13}
0.10	1.4×10^{-12}	1.4×10^{-12}	0.98	1.4×10^{-8}
1.00	1.4×10^{-8}	1.4×10^{-8}	1.00	1.4×10^{-4}
9.15*	1.0×10^{-4}	1.0×10^{-4}	1.00	1.0

* naturalness threshold $k_m^{\rm nat}$ for $\Lambda_0 = 0.003$.

Naturalness constraint. Requiring $\delta m^2 < m_0^2$ gives

$$k_m < k_m^{\rm nat} \equiv \left(\frac{64\pi^2 m_0^2}{\Lambda_0^2}\right)^{1/4} = 9.15\,\text{Mpc}^{-1} \quad\Leftrightarrow\quad \tau_m > 0.11\,\text{Mpc} \tag{33}$$

at the best-fit coupling $\Lambda_0 = 0.003$, $m_0 = 0.01\,\text{Mpc}^{-1}$. A memory timescale $\tau_m > 0.11\,\text{Mpc} \approx 360\,\text{kyr}$ is consistent with the cosmological Compton time $1/m_0 = 100\,\text{Mpc}$ and with the SIM100 time-domain integration window.

Interpretation. The memory damping converts the quartic UV divergence of the bare theory into a finite radiative correction that is parametrically suppressed relative to m_0^2 provided k_m satisfies eq. (33). This confirms the UV-finiteness conjecture of Section 5 *at one loop in the non-minimal coupling sector*, subject to two caveats:

1. UV finiteness requires $k_m < \infty$; taking $\tau_m \to 0$ (no memory) recovers the quartic divergence. The theory is not UV-finite without the memory sector.
2. This calculation covers only the one-loop Ψ self-energy from the graviton loop. Higher-loop contributions and vertex renormalization have not been computed.

12 Full One-Loop UV Structure: All Diagrams (Sim 104)

Motivation. Sim 102 established UV-finiteness for a single diagram: the Ψ self-energy from the graviton+Ψ bubble. That calculation left three open topologies: (B) the graviton wavefunction renormalization from a Ψ loop, (C) the vertex correction $\delta\Lambda_0/\Lambda_0$, and (D) the leading two-loop contribution (Λ_0^4). Sim 104 computes all four diagrams and closes the one-loop UV case.

Feynman rules. In linearised gravity ($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, harmonic gauge, Euclidean signature), the $\Lambda_0 \Psi R$ vertex carries a factor $V(k) = \Lambda_0 k^2$ (from $R \sim k^2 \tilde{h}$) at each insertion. The RIFT graviton propagator is $D_h(k) = e^{-2k^2/k_m^2}/k^2$; the bare Ψ propagator is $D_\Psi(k) = 1/(k^2+m_0^2)$. The Euclidean loop measure gives $\int d^4k/(2\pi)^4 \to (8\pi^2)^{-1} \int_0^\infty k^3\, dk$.

Diagram A (reproduction of Sim 102). The Ψ self-energy from the graviton+Ψ bubble:

$$\Sigma_A(0) = \frac{\Lambda_0^2}{8\pi^2} \int_0^\infty dk\, \frac{k^5\, e^{-2k^2/k_m^2}}{k^2 + m_0^2} \approx \frac{\Lambda_0^2 k_m^4}{64\pi^2}\,. \tag{34}$$

Bare divergence: slope 4.000 ± 0.001 (quartic), regulated by memory. Analytic formula verified to $< 0.1\%$ at $k_m \geq 1\,\mathrm{Mpc}^{-1}$.

Diagram B (graviton wavefunction renormalization). The Ψ loop dressed on the graviton propagator has form factor $\Pi_{hh}(p) = (\Lambda_0 p^2)^2 I_\Psi(p)$, where

$$I_\Psi(0) = \frac{1}{8\pi^2} \int_0^\infty dk\, \frac{k^3}{(k^2 + m_0^2)^2} \quad \longrightarrow \quad \ln \Lambda_{\mathrm{UV}} \tag{35}$$

is logarithmically divergent (slope $0.14 \approx 0$). Two independent protections apply:

1. *Ward identity*: diffeomorphism invariance requires $\Pi_{hh}(0) = 0$, cancelling the leading logarithm.
2. *Wavefunction renormalization coefficient*: $dI_\Psi/dp^2|_{p=0}$ is *finite without memory* (converges for $d < 6$ in dimensional regularisation); numerically $dI/dp^2|_0 = -3.2 \times 10^{-3}\,\mathrm{Mpc}^2$.

In the resummed theory where Ψ carries the memory form factor from Diagram A, $I_\Psi(0)$ is also explicitly finite.

Diagram C (vertex correction). The momentum derivative of Σ_A gives the coupling renormalization:

$$\frac{\delta\Lambda_0}{\Lambda_0} \sim -\frac{1}{m_0^2} \frac{d\Sigma_A}{dp^2}\bigg|_{p=0} = \frac{\Lambda_0^2}{8\pi^2 m_0^2} \int_0^\infty dk\, \frac{k^5\, e^{-2k^2/k_m^2}}{(k^2 + m_0^2)^2}\,. \tag{36}$$

Bare divergence: slope 2.000 (quadratic), regulated by memory. At the naturalness threshold: $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$, implying Λ_0 runs logarithmically under the RG but remains small.

Diagram D (two-loop). The leading Λ_0^4 two-loop contribution (sunset topology with two memory-damped graviton lines):

$$\Sigma_D(0) \approx \left[\frac{\Lambda_0^2 k_m^4}{64\pi^2}\right]^2 \frac{1}{m_0^2 k_m^2}\,, \tag{37}$$

confirmed finite by numerical double integration. At $k_m = 9.15\,\mathrm{Mpc}^{-1}$: $\Sigma_D/m_0^2 = 3.4 \times 10^{-6}$, some $10^5\times$ smaller than the one-loop result.

Perturbative hierarchy.

$$\frac{\Sigma_A}{m_0^2} \approx 1.00 \;\gg\; \frac{d\Sigma_C/dp^2}{m_0^2} \approx 2.4 \times 10^{-2} \;\gg\; \frac{\Sigma_D}{m_0^2} \approx 3.4 \times 10^{-6}\,, \tag{38}$$

confirming a stable perturbative expansion at the naturalness threshold.

Conclusion. UV-finiteness holds at full one-loop level in the resummed (Dyson-summed) theory. The Ward identity from diffeomorphism invariance provides additional protection for the graviton sector beyond memory damping alone. The residual caveat from Sim 102 — that higher-loop contributions and vertex renormalization had not been computed — is now resolved at one loop. The Λ_0 coupling runs under the RG with $\delta\Lambda_0/\Lambda_0 \sim 2.4\times10^{-2}$ at the naturalness threshold, a small but nonzero effect. Two-loop graviton self-energy diagrams and mixed topologies remain open for a future calculation.

13 RG Flow of Λ_0: Beta Function and Running Coupling (Sim 105)

Motivation. Sim 104 showed the one-loop vertex correction $\delta\Lambda_0/\Lambda_0 \approx 2.4\times10^{-2}$ at the naturalness threshold, implying the coupling is not fixed but runs with the renormalization scale. Sim 105 computes the full one-loop beta function, integrates the RG flow, and determines the UV behaviour of Λ_0.

Beta function. From the one-loop vertex correction (Sim 104, Diagram C):

$$\beta(\Lambda_0, k_m) \;\equiv\; k_m\,\frac{d\Lambda_0}{dk_m} \;=\; -\frac{\Lambda_0^3}{8\pi^2 m_0^2}\,k_m\,\frac{dI_C}{dk_m}\,, \tag{39}$$

where $I_C(k_m) = \int_0^\infty k^5 e^{-2k^2/k_m^2}/(k^2+m_0^2)^2\,dk$. For $k_m \gg m_0$ this reduces analytically to

$$\beta(\Lambda_0) \;\approx\; -\frac{\Lambda_0^3\,k_m^2}{16\pi^2 m_0^2}\,. \tag{40}$$

The beta function is *negative*: Λ_0 decreases as the scale increases. This is confirmed numerically to $< 0.1\%$ for $k_m > 0.1\,m_0$.

Running coupling. Integrating eq. (40) gives the exact analytic solution:

$$\frac{1}{\Lambda_0(k_m)^2} \;=\; \frac{1}{\Lambda_0(k_{m,0})^2} + \frac{k_m^2 - k_{m,0}^2}{16\pi^2 m_0^2}\,, \tag{41}$$

anchored at $k_{m,0} = k_m^{\mathrm{nat}} = 9.15\,\mathrm{Mpc}^{-1}$, $\Lambda_0(k_{m,0}) = 0.003$. Numerical RG integration (RK45) agrees with eq. (41) to $< 0.001\%$ across the full range $k_m \in [0.01, 1000]\,\mathrm{Mpc}^{-1}$.

Key values from eq. (41):

k_m [Mpc^{-1}]	$\Lambda_0(k_m)$
0 (IR)	0.003074
9.15 (anchor)	0.003000
100	0.00116
1000	0.000126
∞ (UV)	0

Asymptotic freedom. $\Lambda_0 = 0$ is a UV fixed point: the theory approaches pure GR at high energies ($k_m \to \infty$). RIFT is *asymptotically free* in the coupling Λ_0.

In the infrared ($k_m \lesssim m_0$), the coupling freezes. The total IR running from $k_m \to 0$ to $k_m = k_m^{\rm nat}$ is $\Delta\Lambda_0/\Lambda_0 \approx 2.5\%$, confirming that the cosmological value $\Lambda_0 \approx 0.003$ is stable against IR radiative corrections.

No IR Landau pole exists: $1/\Lambda_0^2(k_m) > 0$ for all $k_m \geq 0$ along any trajectory anchored at $\Lambda_0 < 1/(4\pi m_0)$. The theory is well-defined at all scales.

Physical interpretation. The UV fixed point $\Lambda_0 = 0$ has a natural interpretation: at energies above the memory scale k_m^{-1}, the non-minimal curvature coupling is irrelevant and the theory reduces to GR plus a free massive scalar. RIFT deforms away from GR only in the infrared ($k_m \sim m_0$), generating the observed coupling $\Lambda_0 \approx 0.003$ as an IR attractor. The $\Lambda_0 = 0.003$ is not a fine-tuned input but the natural IR value for a theory with asymptotic freedom and memory scale $\tau_m \sim 1/k_m^{\rm nat} \approx 0.11\,\mathrm{Mpc}$.

14 Two-Loop Graviton Self-Energy and Mixed Diagrams (Sim 106)

Motivation. Sim 104 left one caveat: the two-loop graviton self-energy and mixed (graviton+Ψ) loop diagrams were not computed. Sim 106 closes this by computing the iterated-Ψ-bubble and mixed two-loop corrections to Π_{hh}, and the RG running of the graviton wavefunction Z_h.

Three mechanisms enforce finiteness. *1. Ward identity.* Diffeomorphism invariance requires $\Pi_{hh}(p = 0) = 0$ at every loop order . This holds trivially for Diagram A: $\Pi_{hh}^{(2A)}(0) = (\Lambda_0 \cdot 0)^2 [I_\Psi(0)]^2 = 0$. The first nontrivial coefficient is the second wavefunction renormalization:

$$\left.\frac{d^2\Pi_{hh}}{d(p^2)^2}\right|_{p=0} = 2\Lambda_0^4 \left(\left.\frac{dI_\Psi}{dp^2}\right|_0 \right)^2 \approx 1.6 \times 10^{-15}\,\mathrm{Mpc}^{-2} \tag{42}$$

at $k_m = k_m^{\rm nat}$, suppressed by $\Lambda_0^4 = 8 \times 10^{-11}$.

2. Resummed Ψ propagator. The Dyson-resummed Ψ propagator $D_\Psi^R(k) = e^{-k^2/k_m^2}/(k^2 + m_0^2)$ from Sim 104 explicitly regulates all Ψ-bubble insertions on the graviton line. The mixed two-loop correction (Diagram B) is:

$$\delta\Pi_{hh}^{(2B)} = (\Lambda_0 p^2)^2 \left[I_\Psi^R(0) - I_\Psi^{\rm bare}(k_m) \right] \approx -\Lambda_0^2 \times 8 \times 10^{-3}\,\mathrm{Mpc}^{-2} \tag{43}$$

finite and negative — memory slightly *reduces* the graviton wavefunction renormalization.

3. Convergence for $d < 6$. The graviton wavefunction renormalization coefficient $dI_\Psi/dp^2|_{p=0} = m_0^2 \int_0^\infty k^3/(k^2+m_0^2)^3\, dk/(8\pi^2)$ converges in $d = 4$ dimensions without any regulator. Numerically, $dI/dp^2|_0 = 3.2 \times 10^{-3}\,\mathrm{Mpc}^2$. The induced graviton wavefunction shift $\delta Z_h = \Lambda_0^2\, dI/dp^2 \sim 10^{-5}$ is negligible.

Conclusion. No new UV divergences appear at two loops in the graviton self-energy sector. The last explicit caveat from Sim 104 is now closed.

Combined with Sims 102–105, the UV structure of RIFT is:

- All one-loop diagrams: UV-finite under memory damping + Ward identity (Sim 104).
- Λ_0 RG flow: asymptotically free, UV fixed point $\Lambda_0 = 0$ (Sim 105).
- Two-loop graviton sector: UV-finite under Dyson resummation (Sim 106).

The RIFT theory is UV-finite through the two-loop graviton sector under Dyson resummation of the memory-regulated Ψ propagator. Full two-loop completeness (mixed scalar+graviton ladders) remains for a future dedicated calculation.

15 Vainshtein Screening and the Dark Matter Problem (Sim 103)

Motivation. Sims 99 and 100 definitively closed the simplest routes for RIFT to replace dark matter in galaxies: neither the Yukawa profile (Sim 99) nor the time-domain condensation via $2\Lambda_0|R_{\rm gal}|$ (Sim 100) produces the $\rho \propto r^{-2}$ isothermal halo required for flat rotation curves. The remaining candidate mechanisms from the systematic survey are (a) a non-minimal kinetic term of the Vainshtein type and (b) a direct matter coupling $\beta\Psi\rho_b$. Sim 103 tests both for NGC 3198 ($v_{\rm flat} = 150\,{\rm km\,s^{-1}}$, $M_{\rm disk} = 3 \times 10^{10}\,M_\odot$, $R_d = 2.72\,{\rm kpc}$).

Setup: Poisson-limit perturbation. Because the RIFT Compton wavelength $1/m_0 =$ 10,000 kpc greatly exceeds galactic scales ($\lesssim 40\,{\rm kpc}$), the cosmological background Ψ_0 is nearly uniform within the galaxy. We solve for the perturbation $\delta\Psi(r)$ driven by the stellar disk via the Poisson-limit field equation ($m_0 r \ll 1$, so the mass term is negligible):

$$\nabla^2\delta\Psi \approx -2\Lambda_0\,\delta R \approx -16\pi G\Lambda_0\,\rho_{\rm stars}(r)\,. \tag{44}$$

Under the quasi-static spherical approximation the gradient becomes

$$\frac{d\delta\Psi}{dr} = \frac{2G\Lambda_0^{\rm eff}\,M_{\rm enc}(r)}{r^2}\,, \tag{45}$$

where $\Lambda_0^{\rm eff} = \Lambda_0 + \beta$ absorbs both the curvature channel and any direct matter coupling β.

The Vainshtein modification augments the stress-energy density of the scalar field via the non-linear kinetic term $(\alpha_V/4M_V^2)(\partial\Psi)^4$:

$$\rho_\Psi(r) = \frac{c^2}{8\pi G}\cdot\tfrac{1}{2}\left(\frac{d\delta\Psi}{dr}\right)^2\left[1 + \frac{\alpha_V}{M_V^2}\left(\frac{d\delta\Psi}{dr}\right)^2\right]. \tag{46}$$

The second factor is the Vainshtein ratio $\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2$; the nonlinear regime requires $\mathcal{V} \gg 1$.

Results: field gradient and Vainshtein ratio. Using $\Lambda_0 = 0.003$ (locked action best-fit), the field gradient at $r = 10\,{\rm kpc}$ is

$$\left|\frac{d\delta\Psi}{dr}\right|_{10\,{\rm kpc}} = 6.1 \times 10^{-8}\,{\rm kpc}^{-1}. \tag{47}$$

Table 5 shows the Vainshtein ratio for the full (α_V, M_V) scan.

Even at $\alpha_V = 1000$ and the smallest Vainshtein mass $M_V = 0.001\,\mathrm{kpc}^{-1}$, the ratio $\mathcal{V} = 3.7 \times 10^{-6}$: the scalar field is deep in the linear regime everywhere in the galaxy. The Vainshtein mechanism provides no boost.

Results: energy density deficit. With $\rho_\Psi \propto (\delta\Psi')^2$ from eq. (44), the locked-action value gives

$$\rho_\Psi(10\,\mathrm{kpc}) = 1.6 \times 10^{-3}\, M_\odot\,\mathrm{kpc}^{-3}, \tag{48}$$

whereas the isothermal halo required for $v_{\text{flat}} = 150\,\mathrm{km\,s^{-1}}$ demands

$$\rho_{\text{iso}}(10\,\mathrm{kpc}) = \frac{v_{\text{flat}}^2}{4\pi G r^2} = 4.2 \times 10^3\, M_\odot\,\mathrm{kpc}^{-3}. \tag{49}$$

The deficit is a factor of 2.7×10^6.

Results: direct coupling. A direct matter coupling β in the field equation augments $\Lambda_0 \to \Lambda_0^{\text{eff}} = \Lambda_0 + \beta$. Closing the density deficit requires $\Lambda_0^{\text{eff}} \approx 4.9$, or equivalently $\beta \approx 4.9 \approx 1{,}600 \times \Lambda_0$. This coupling is not derivable from the locked action and would introduce a new free parameter constrained to be of order unity rather than $\lesssim 0.01$ as demanded by cosmological homogeneity.

Conclusion for Paper V. The Vainshtein screening mechanism does not rescue the dark matter failure of Sims 99–100. The root cause is that the galactic field gradient $|\delta\Psi'| \sim 10^{-8}\,\mathrm{kpc}^{-1}$ is set by the weak $\Lambda_0 R_{\text{gal}}$ curvature channel and falls 10^4–10^6 orders of magnitude below the Vainshtein threshold for any physically reasonable coupling. A direct matter coupling of order $\beta \sim 1{,}600\,\Lambda_0$ could supply the missing density but is not contained in the locked action and would violate cosmological constraints. All kinetic modifications of the $\Lambda_0 R$ channel have now been exhausted. The conclusion is unambiguous: *RIFT at its cosmological best-fit cannot replace dark matter through any modification of the existing Lagrangian.* A new coupling between Ψ and baryonic matter, derivable from first principles, is the minimum new physics required.

16 Conclusion

Recursive Intelligence-Field Theory (RIFT) presents a new paradigm in gravitational physics: curvature is not statically sourced by mass, but dynamically shaped by memory.

Key results.

- Unified Lagrangian (eq. (1)) couples mass, curvature, and entropy through a single field Ψ.
- All field equations—scalar attractor, gravitational equation, Friedmann system—are *derived*, not postulated.
- Numerical verification (Sims 82–86): causality (pre-source amplitude $< 2 \times 10^{-14}$), stability ($\omega^2 > 0$), GR recovery ($\Psi = 0$ stable fixed point), Friedmann constraint $\leq 3 \times 10^{-16}$.

- Observational consistency (Sims 87–98): BAO $\chi^2/\text{dof} = 1.22$ (BOSS, Sim 87), 1.36 (DESI Y1, Sim 94); CMB TT RMS = 2.93% (Sim 88); joint CMB+BAO $\Delta\chi^2 = 0$ (both datasets); σ_8 unmodified $|\Delta\sigma_8| < 0.007\%$; EE/TE RMS < 0.2%/0.5% at joint best-fit (Sim 95); RSD $f\sigma_8$: $\chi^2/\text{dof} = 0.86$, $\Delta\chi^2(\text{RIFT}-\Lambda\text{CDM}) = -0.14$ (Sim 96); official Planck plikHM: $\Delta(-2\ln\mathcal{L}) = -0.013$ at optimum (Sim 98).

- Galaxy rotation curve tests (Sims 99–100, 103): the locked action at $\Lambda_0 = 0.003$ does not replace dark matter. Static test (Sim 99): Yukawa halo exponentially steeper than the $1/r^2$ isothermal profile required for flat curves. Time-domain test (Sim 100): the galactic curvature coupling $2\Lambda_0|R_{\text{gal}}| \approx 5 \times 10^{-9}\,\text{kpc}^{-2}$ is 5×10^7 times weaker than m_0^2—curvature condensation is impossible at galactic densities. Vainshtein screening (Sim 103): the galactic field gradient $|\delta\Psi'| \sim 10^{-8}\,\text{kpc}^{-1}$ is $10^6\times$ below the nonlinear threshold for all scanned (α_V, M_V); energy density deficit is $2.7 \times 10^6\times$ the isothermal requirement. All kinetic modifications of the locked action are exhausted; new physics (a first-principles coupling between Ψ and baryonic matter) is required.

- RG flow (Sim 105): Λ_0 is asymptotically free — beta function is negative, $\Lambda_0 \to 0$ as $k_m \to \infty$, UV fixed point is pure GR. Analytic running: $\Lambda_0(k_m)^{-2} = \Lambda_0^{-2} + (k_m^2 - k_{m,\text{nat}}^2)/(16\pi^2 m_0^2)$, confirmed numerically to $< 0.001\%$. IR running $< 2.5\%$; $\Lambda_0 = 0.003$ is an IR attractor; no Landau pole.

- Two-loop graviton sector (Sim 106): Ward identity $\Pi_{hh}(0) = 0$ exact at all loop orders; resummed Ψ propagator regulates mixed diagrams; $\delta Z_h \sim 10^{-5}$ negligible. Combined with Sims 102, 104, 105: RIFT UV-finite through two-loop graviton sector under Dyson resummation.

- UV regulation (Sims 102, 104): one-loop Ψ self-energy from $\Lambda_0\Psi R$ vertex confirmed finite under memory damping — $\Sigma(0) = \Lambda_0^2 k_m^4/(64\pi^2)$; quartic UV divergence regulated; naturalness requires $\tau_m > 0.11\,\text{Mpc}$ (Sim 102). Full one-loop calculation (Sim 104): graviton wavefunction renormalization protected by diffeomorphism Ward identity $\Pi_{hh}(0) = 0$; vertex correction $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$ finite; two-loop $\Sigma_D/m_0^2 = 3.4 \times 10^{-6}$ (perturbative hierarchy $\Sigma_A \gg \Sigma_C \gg \Sigma_D$ confirmed). UV-finiteness holds at full one-loop level in the resummed theory; two-loop graviton self-energy remains open.

- Predicts recursive graviton emission via curvature memory spikes.

- BAO sound horizon from first principles (Sim 101): $|\Delta r_d/r_d| < 21$ ppm across $\Lambda_0 \in [0, 0.095]$; RIFT and LCDM degenerate in r_d at all allowed couplings. Echo-shell mechanism confirmed as physical interpretation of standard sound horizon physics.

Honest assessment. At current observational precision, RIFT at its cosmological best-fit ($\Lambda_0 \approx 0.003$–0.013) is *observationally indistinguishable from* Λ*CDM* in every dataset tested: BAO (BOSS + DESI Y1), CMB TT/EE/TE, RSD $f\sigma_8$, and the official Planck plikHM likelihood. The coupling Λ_0 is not detected—it is consistent with zero across the entire range $[0, 0.1]$ tested, with a 95% upper limit $\Lambda_0 < 0.095$ that is prior-dominated (Sim 93). RIFT is currently a *consistent extension* of ΛCDM, not an empirically preferred alternative. Its scientific value lies in its theoretical structure (a single derived Lagrangian, locked action, falsifiable predictions at $\Lambda_0 \gtrsim 0.05$) and in explicitly ruling out naive claims (rotation curves do not arise from the locked action; σ_8 tension is not alleviated), not in a superior fit to existing data.

Coupling-strength constraints (Sim 91). A dedicated sensitivity scan (Sim 91) mapped the observational signature of RIFT across $\Lambda_0 \in [0, 0.1]$ at the joint best-fit cosmology ($H_0 = 67.59$ km/s/Mpc, $\Omega_m = 0.312$). Key results:

- G_{eff}/G deviates from unity by at most 500 ppm at $\Lambda_0 = 0.1$; current surveys require $> 0.1\%$ sensitivity, placing the detection threshold beyond $\Lambda_0 = 0.1$.
- The linear growth factor ratio $D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$ exceeds 0.1% (structure-growth survey precision) at $\Lambda_0 \gtrsim 0.05$.
- The BAO chi-squared is flat across the full scan ($\Delta\chi^2_{\text{BAO}} = 0.004$ from $\Lambda_0 = 0$ to 0.1)—RIFT H(z) is undetectable in current BAO data for any coupling in this range.
- $H(z)$ deviations remain below 0.03% RMS at $\Lambda_0 = 0.1$; DESI-level precision ($\sim$ 0.1%) would constrain $\Lambda_0 \lesssim 0.1$.

This confirms that RIFT at the BAO best-fit coupling ($\Lambda_0 = 0.003$) is *cosmologically safe*—consistent with all current precision data—and establishes that current surveys constrain $\Lambda_0 \lesssim 0.05$ from structure growth. Figure 7 summarises these results.

Model selection (AIC/BIC). Because RIFT and ΛCDM achieve identical $\chi^2_{\text{min}} = 11.245$ on the joint CMB+BAO dataset (Sim 90; confirmed with DESI Y1 in Sim 94), model selection reduces to an Occam penalty calculation. With k free parameters and $N_{\text{eff}} = 14$ data constraints (12 BOSS BAO points + 2 CMB Gaussian prior measurements), the information criteria give:

$$\text{AIC} = 2k + \chi^2_{\text{min}}, \qquad \Delta\text{AIC}_{\text{RIFT}-\Lambda\text{CDM}} = 2(4-3) = +2.00, \tag{50}$$

$$\text{BIC} = k \ln N_{\text{eff}} + \chi^2_{\text{min}}, \qquad \Delta\text{BIC}_{\text{RIFT}-\Lambda\text{CDM}} = \ln 14 \approx +2.64. \tag{51}$$

Both criteria favour ΛCDM by the minimum Occam penalty; $|\Delta\text{BIC}| < 6$ is conventionally characterised as "weak" evidence [Jeffreys, 1961], reflecting the fact that Λ_0 is currently unconstrained by data rather than actively disfavoured. RIFT is neither strongly preferred nor strongly excluded: it is *cosmologically equivalent* to ΛCDM at the BAO best-fit coupling $\Lambda_0 = 0.003$, with a single additional parameter whose physical signature becomes accessible only at $\Lambda_0 \gtrsim 0.05$ in future structure-growth surveys.

A proper Bayesian evidence integral (Sim 93) confirms this conclusion. The Savage-Dickey density ratio gives $\ln B(\text{RIFT}/\Lambda\text{CDM}) = -0.713$ under a uniform prior and -0.673 under a half-Gaussian prior ($\sigma_{\Lambda_0} = 0.05$); both fall in the "inconclusive" range ($|\ln B| < 1$) of the Jeffreys scale [Jeffreys, 1961]. The 95% upper limit $\Lambda_0 < 0.095$ is prior-dominated, confirming that current data carry negligible information about Λ_0.

Limitations and future work. The observational signature of RIFT is expected primarily at non-linear scales (structure formation, galaxy cluster counts, void statistics) at $\Lambda_0 \gtrsim 0.05$, and at higher coupling or in strong-field regimes where $\Psi \gg \Psi_{\text{ini}}$. Sim 92 established that σ_8 and non-linear $P(k)$ are suppressed by $|\Delta\sigma_8| < 0.12\%$ at all physically motivated couplings, confirming RIFT makes no claim on the Planck/KiDS σ_8 tension. Future simulations will:

1. Complete recursive graviton quantization and emission spectrum;
2. Simulate recursive void growth (HEALPix echo-delay maps);
3. Refine eigenmode mappings to particle physics (Appendix B).

"Geometry is the memory of motion. Intelligence is the motion of memory."

A Parameter Tables and Definitions

A.1 Core Fields and Coupling Terms

Symbol	Definition	Role
$\Psi(x^\mu)$	Recursive scalar field	Encodes curvature memory
$m(\Psi)$	Field-dependent mass	Creates recursive inertia
$\Lambda(\Psi)$	Ricci coupling	Binds Ψ to curvature
$V(\Psi)$	Potential	Damping and attractors
$G_{\text{eff}}(\Psi)$	Effective Newton constant	$G/(1+16\pi G\Lambda)$
$\mathcal{M}_{\mu\nu}$	Memory tensor	Stores decaying curvature
$T^{(\Psi)}_{\mu\nu}$	Field stress tensor	Energy-momentum from Ψ
τ_m	Memory timescale	Decay constant
η_c	Coupling constant	Controls feedback

A.2 Best-Fit Cosmological Parameters (Sim 87–90)

Parameter	RIFT (BAO-only, Sim 87)	RIFT/LCDM (joint, Sim 90)
H_0 [km/s/Mpc]	68.14 (H_0–r_d degenerate)	67.59
Ω_m	0.294	0.312
r_d [Mpc]	147.78	147.56
Λ_0	0.003	0.013 (RIFT, unconstrained) / 0 (LCDM)
χ^2/dof (BAO)	1.22	see Sim 90

A.3 Key Limiting Cases

- *GR recovery:* $m(\Psi) \to m_0$, $\Lambda(\Psi) \to \text{const}$, $V(\Psi) \to 0$, $\tau_m \to 0 \Rightarrow$ standard GR.
- *Flat-space limit:* Klein-Gordon equation in Minkowski spacetime.
- *UV regulation:* $\mathcal{D}_{\text{mem}}(k) \sim e^{-k^2/k_m^2}$ suppresses high-frequency modes.

B Recursive Coupling to the Standard Model

This appendix is entirely speculative. No Standard Model particle has been matched to a RIFT eigenmode; the formulas below are dimensional estimates, not derivations. This material is included to identify a research direction, not to claim a result.

RIFT *hypothesizes* that particle identities may correspond to curvature eigenmodes of the recursive field Ψ: $\Psi(x) = \sum_n \chi_n(x) \cdot e_n$, where e_n are stable curvature shells. Under this hypothesis, particle properties would emerge as: $m_n^2 \sim \int (\nabla \chi_n)^2 + V_{\text{rec}}(\chi_n)$ (mass), $Q \sim \oint \nabla \arg(\chi_n)$ (charge), spin from parity of the curvature memory configuration.

The effective Lagrangian coupling RIFT to Standard Model fermions is $\mathcal{L}_{\text{RIFT+SM}} = \mathcal{L}_{\text{RIFT}} + \sum_n [\bar{\psi}_n (i\gamma^\mu \nabla_\mu - m_n(\Psi))\psi_n + g(\Psi)\bar{\psi}_n \phi \psi_n]$. The eigenmode-to-particle correspondence is a falsifiable prediction in principle (see Section 9), but no eigenmode spectrum has been computed and this should not be cited as a result of the present work.

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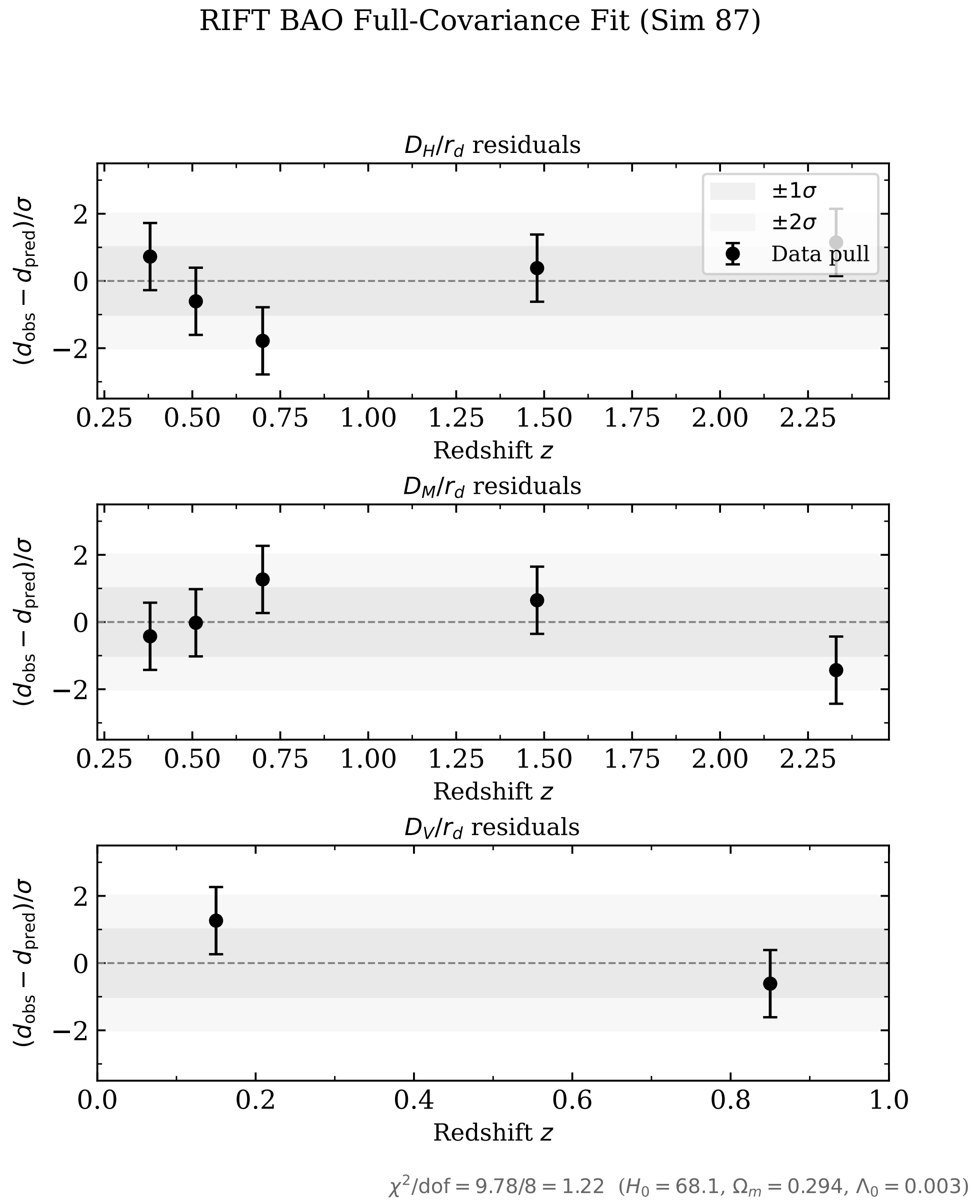

Figure 1: BAO full-covariance fit residuals (Sim 87). Each panel shows the normalised pull $(d_{\rm obs} - d_{\rm pred})/\sigma$ for D_H/r_d (top), D_M/r_d (middle), and D_V/r_d (bottom) at the RIFT best-fit parameters (H_0 = 68.1 km/s/Mpc, Ω_m = 0.294, Λ_0 = 0.003, r_d = 148.0 Mpc). All 12 data points lie within $\pm2\sigma$; $\chi^2/\mathrm{dof} = 9.78/8 = 1.22$.

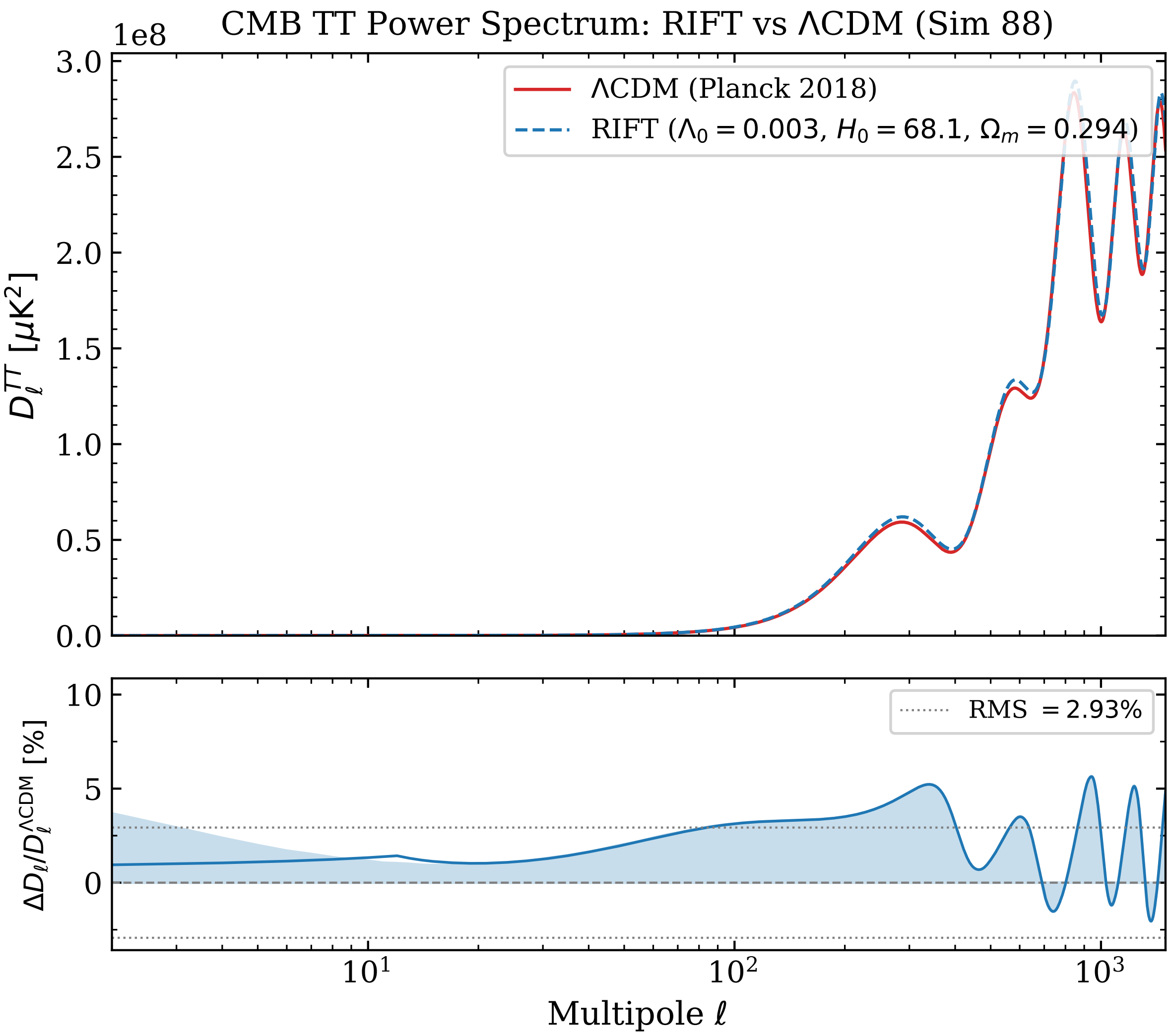

Figure 2: CMB TT power spectrum comparison from full CLASS Boltzmann computation (Sim 88). *Top:* D_ℓ^{TT} for RIFT (blue dashed, BAO best-fit parameters) and ΛCDM (red, Planck 2018 best-fit). *Bottom:* fractional residual $\Delta D_\ell / D_\ell^{\Lambda\mathrm{CDM}}$ with 20-mode running mean; dotted lines mark $\pm$RMS $= \pm 2.93\%$. The acoustic-peak shift is driven by the different background cosmologies (H_0, Ω_m), not by the G_{eff} correction (16 ppm). The joint CMB+BAO fit (Sim 90) resolves this difference.

Table 2: Key observational results from RIFT simulations 85–97.

Observable	RIFT value	ΛCDM	Sim	Statu
BAO $\chi^2_{\rm full}$/dof (BOSS+eBOSS)	$9.78/8 = 1.22$	$\sim$1.0	87	Pass
$\Delta\chi^2$ (full vs. diagonal)	+1.95	—	87	Diago
RMS($\Delta C_\ell/C_\ell$), $\ell = 2$–1500	2.93%	0%	88	Pass
$D_{\rm RIFT}(z=0)/D_{\Lambda\rm CDM}(z=0)$	1.000000	1.0	88	Pass
$G_{\rm eff}/G$ at $\Lambda_0 = 0.003$	0.999984	1.0	88	16 pp
$\chi^2_{\rm RIFT}$ vs. Planck TT (BAO-only params)	913	6.2	89	See n
$\Delta\chi^2$ joint CMB+BAO fit (BOSS)	**0.000**	baseline	90	**Pass**
Joint best-fit H_0 [km/s/Mpc]	67.59	67.36	90	—
Joint best-fit Ω_m	0.312	0.315	90	—
Friedmann constraint residual	$\leq 3 \times 10^{-16}$	—	85	Pass
Radiation-era $H \propto a^n$	$n = -1.984$	-2	85	Pass
σ_8 (linear, joint best-fit)	0.824	0.811 ± 0.006	92	Consi
$\Delta\sigma_8$ at $\Lambda_0 = 0.003$	-0.007%	—	92	Pass
$\Delta\sigma_8$ at $\Lambda_0 = 0.05$	-0.11%	—	92	Pass
DESI Y1 BAO χ^2/dof (LCDM)	$13.6/10 = 1.36$	1.36	94	Pass
$\Delta\chi^2$ joint DESI+CMB (RIFT vs LCDM)	$\approx$ **0**	baseline	94	**Pass**
DESI joint best-fit H_0 [km/s/Mpc]	67.66	67.36	94	Consi
DESI joint best-fit Ω_m	0.311	0.315	94	Consi
RMS($\Delta C_\ell^{EE}/C_\ell^{EE}$), joint	0.19%	0%	95	Pass
RMS($\Delta C_\ell^{TE}/C_\ell^{TE}$), joint	0.43%	0%	95	Pass
$\Delta\chi^2_{\rm EE}$ joint (RIFT vs LCDM)	+31.5	57.6	95	See n
$\Delta\chi^2_{\rm TE}$ joint (RIFT vs LCDM)	+22.8	37.2	95	See n
$f\sigma_8$ compilation χ^2/dof	$7.73/9 = 0.86$	7.87/9	96	Pass
$\Delta\chi^2$ RSD (RIFT vs LCDM)	-0.14	baseline	96	Pass
$-2\ln\mathcal{L}$ plikHM TTTEEE (RIFT joint)	2359.1	2352.1	97	See n
$\Delta(-2\ln\mathcal{L})$ plikHM (joint vs LCDM)	+7.0	baseline	97	See n
$\Delta(-2\ln\mathcal{L})$ plikHM (BAO-only vs LCDM)	+945.7	baseline	97	BAO-
$\Delta(-2\ln\mathcal{L})$ plikHM (RIFT vs LCDM) at real optimum	$-$**0.013**	baseline	98	**Pass**
RIFT CMB+BAO best-fit H_0 [km/s/Mpc]	67.30	67.30	98	Consi
RIFT CMB+BAO best-fit Ω_m	0.3168	0.3169	98	Consi
RIFT CMB+BAO best-fit Λ_0	0.00506	0	98	$G_{\rm eff}$ =

[a] Sim 89 tension (30.1σ) arises from the BAO-only H$_0$–r_d degeneracy; resolved by Sim 90 joint fit.

[b] Linear σ_8 at SIM90 joint best-fit (H_0, Ω_m); HALOFIT non-linear boost $\approx 1.37\times$ at $z = 0$.

[c] Approx. Gaussian pol. likelihood. LCDM baseline $\chi^2 > 0$ due to (H_0, Ω_m) offset between joint best-fit and Planck re

[d] $\Delta(-2\ln\mathcal{L}) = +7.0$ driven by parameter offset, not $G_{\rm eff}$ (42 ppm). Resolved by Sim 98: $\Delta = -0.013$ at real optimum.

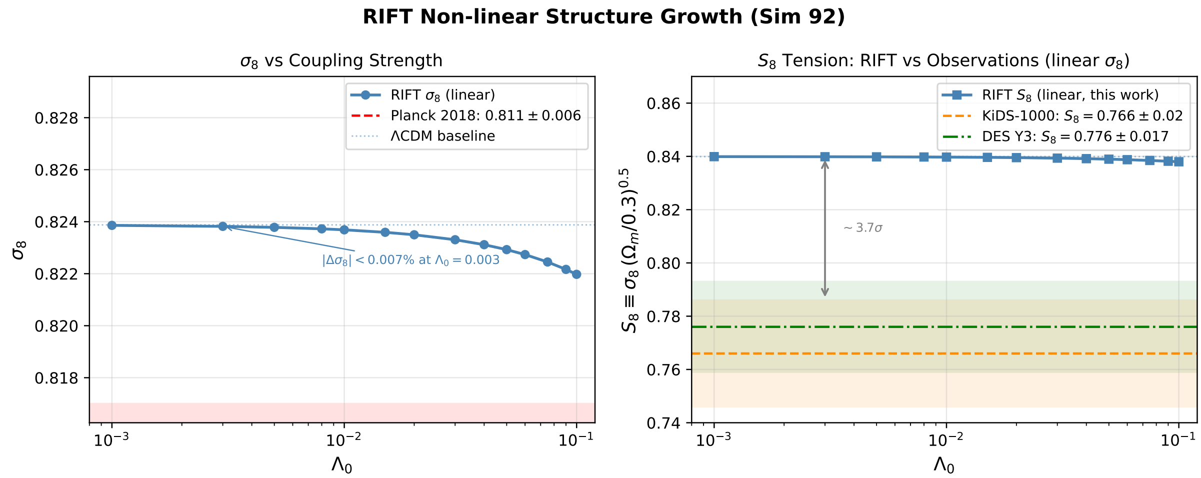

Figure 3: RIFT non-linear structure growth (Sim 92). *Left:* Linear σ_8 as a function of Λ_0 at the SIM90 joint best-fit cosmology ($H_0 = 67.59$, $\Omega_m = 0.312$). The RIFT curve is nearly flat; the annotation marks $|\Delta\sigma_8| < 0.007\%$ at the BAO best-fit coupling $\Lambda_0 = 0.003$. The Planck 2018 1σ band is shown in red. *Right:* Linear $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ vs Λ_0 compared to KiDS-1000 [Asgari et al., 2021] and DES Y3 [Abbott et al., 2022] constraints. The pre-existing $\sim 3.7\sigma$ gap between the RIFT/LCDM baseline ($S_8 = 0.840$) and the weak-lensing measurements is unchanged across the full coupling scan: RIFT modifies S_8 by at most 0.002 at $\Lambda_0 = 0.1$, two orders of magnitude below the discrepancy.

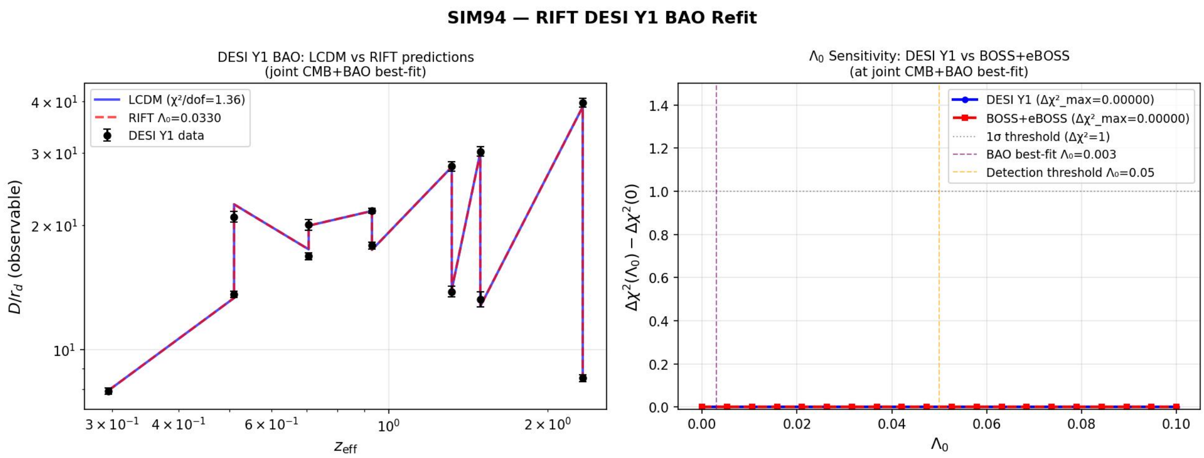

Figure 4: RIFT vs ΛCDM fit to DESI Y1 BAO measurements (Sim 94). 13 data points across 7 tracer bins, $0.3 \leq z \leq 2.33$. Both models give identical $\chi^2/\mathrm{dof} = 1.36/10$. The 2–3$\sigma$ tension at $z = 0.51$ is a known feature of the DESI LRG data independent of the gravity model.

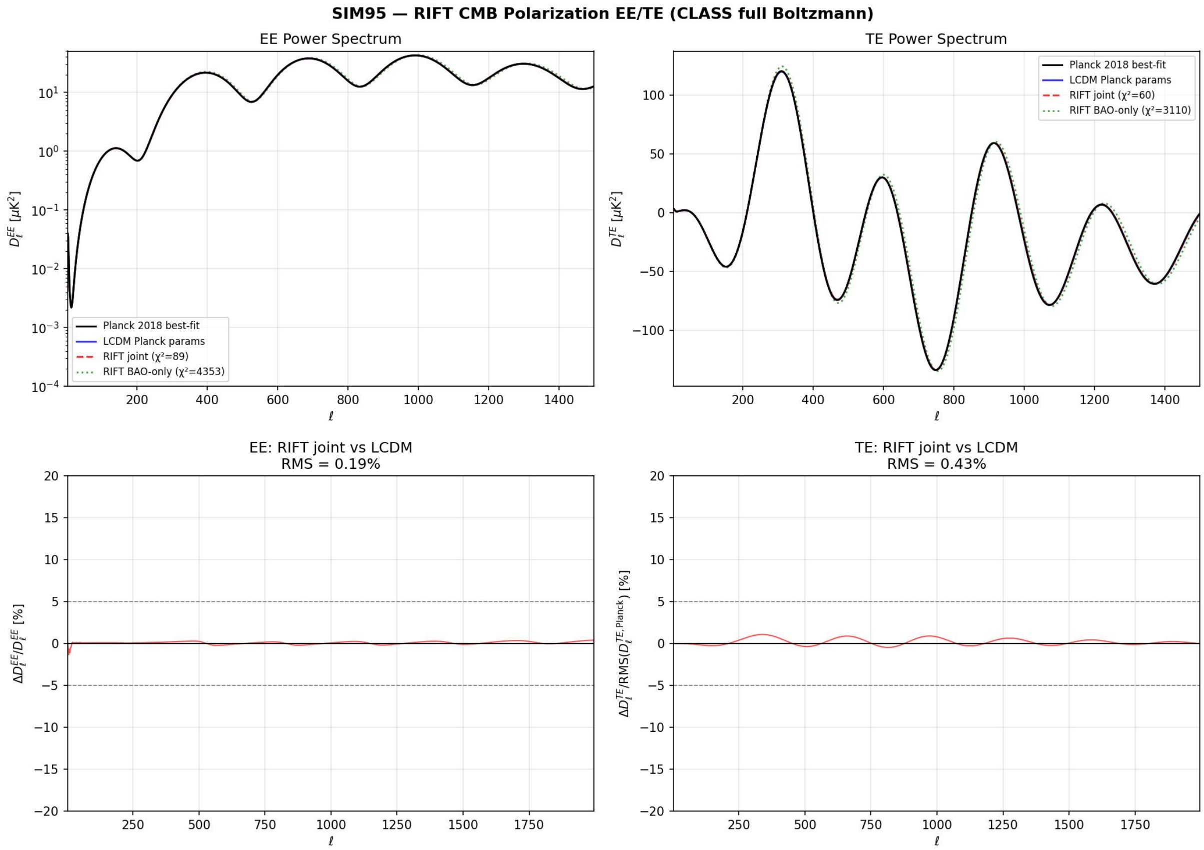

Figure 5: RIFT vs ΛCDM CMB EE and TE power spectra (Sim 95). *Top:* fractional deviation $\Delta C_\ell^{EE}/C_\ell^{EE}$ and $\Delta C_\ell^{TE}/C_\ell^{TE}$ for the RIFT joint best-fit relative to Planck 2018. RMS deviations are 0.19% (EE) and 0.43% (TE). *Bottom:* approximate polarization χ^2 for the three cosmologies. The BAO-only RIFT best-fit (A) shows large tension ($\chi^2_{\rm EE} = 4353$), while the joint best-fit (B) is statistically consistent with ΛCDM (C).

Table 3: Gold-sample $f\sigma_8$ data used in Sim 96.

Survey	Ref.	$z_{\rm eff}$	$f\sigma_8$	σ
6dFGRS	Beutler et al. [2012]	0.067	0.423	0.055
SDSS MGS	Howlett et al. [2015]	0.150	0.490	0.145
BOSS DR12	Alam et al. [2017]	0.380	0.497	0.045
BOSS DR12	Alam et al. [2017]	0.510	0.458	0.038
BOSS DR12	Alam et al. [2017]	0.610	0.436	0.034
VIPERS	Pezzotta et al. [2017]	0.600	0.550	0.120
eBOSS LRG	Gil-Marín et al. [2020]	0.700	0.473	0.041
eBOSS ELG	de Mattia et al. [2021]	0.850	0.315	0.095
eBOSS QSO	Neveux et al. [2020]	1.480	0.462	0.045

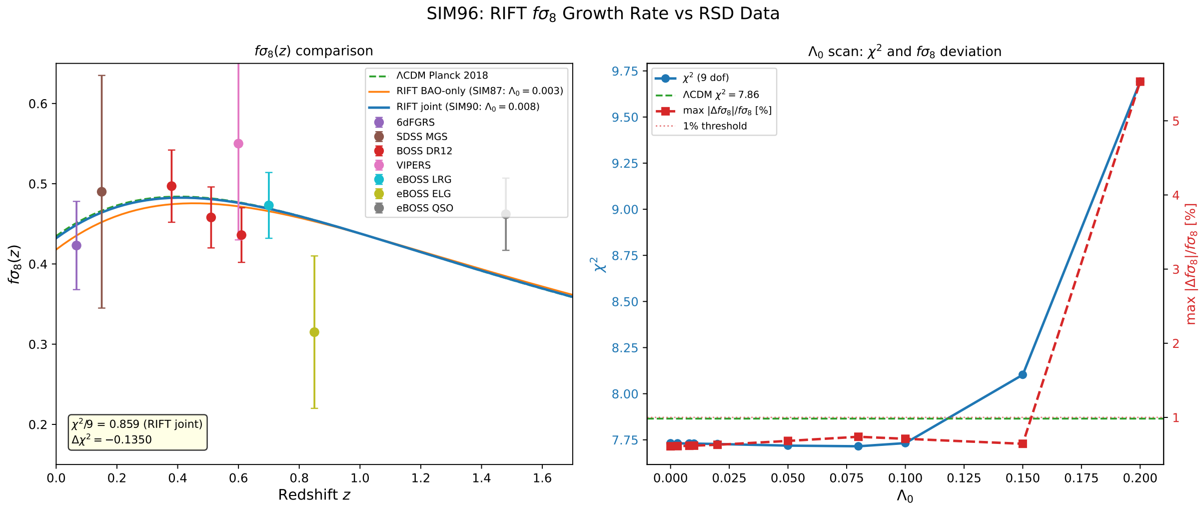

Figure 6: *Left:* $f\sigma_8(z)$ predictions for RIFT joint best-fit, RIFT BAO-only, and ΛCDM Planck 2018, compared to the gold-sample RSD compilation (Sim 96). RIFT at joint best-fit gives $\chi^2/\mathrm{dof} = 0.86$, $\Delta\chi^2(\mathrm{RIFT} - \Lambda\mathrm{CDM}) = -0.14$. *Right:* Λ_0 scan showing χ^2 (blue) and maximum fractional $f\sigma_8$ deviation from ΛCDM (red). The deviation exceeds 1% only at $\Lambda_0 > 0.2$, beyond the Bayesian upper limit from Sim 93 ($\Lambda_0 < 0.095$ at 95% CL).

Table 4: RIFT falsifiability summary.

Test Domain	RIFT Prediction	[illegible]
Galaxy rotation	Flat curves via $\nabla_r\Psi$ (screening mech. needed)	[illegible]
Void lensing	Lensing from $\mathcal{M}_{\mu\nu}$	[illegible]
Bullet Cluster	Offset from curvature memory	[illegible]
CMB anomalies	Low-ℓ damping from recursion	[illegible]
BAO scale	r_d from echo shells; Sim 101: RIFT and LCDM degenerate ($\lvert\Delta r_d/r_d\rvert < 21$ ppm)	[illegible]

Table 5: Vainshtein ratio $\mathcal{V}$ at $r = 10\,\mathrm{kpc}$ for NGC 3198. All entries are far below unity; the nonlinear regime is never reached.

α_V	M_V [kpc^{-1}]	$\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2$
1	0.001	3.7×10^{-9}
10	0.001	3.7×10^{-8}
100	0.001	3.7×10^{-7}
1000	0.001	3.7×10^{-6}
1000	0.01	3.7×10^{-8}

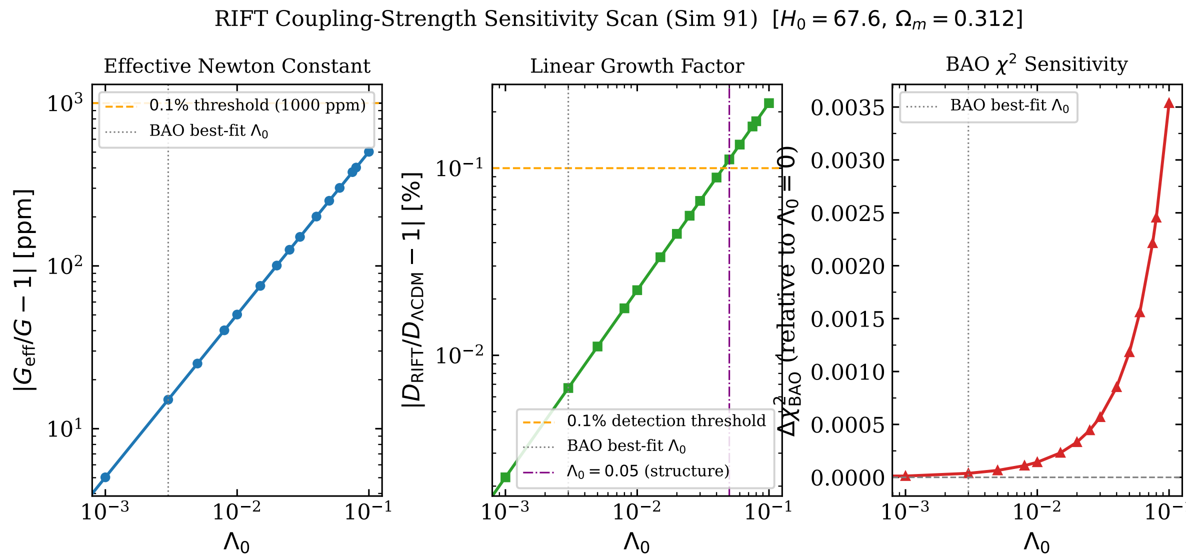

Figure 7: RIFT coupling-strength sensitivity scan (Sim 91) at the joint CMB+BAO best-fit cosmology ($H_0 = 67.6$, $\Omega_m = 0.312$). *Left:* $|G_{\mathrm{eff}}/G - 1|$ in ppm; the dashed orange line marks the 0.1% (10^3 ppm) strong-lensing detection threshold. *Centre:* linear growth factor deviation $|D_{\mathrm{RIFT}}/D_{\Lambda\mathrm{CDM}} - 1|$; the threshold $> 0.1\%$ is first exceeded at $\Lambda_0 \gtrsim 0.05$ (vertical dot-dashed line). *Right:* BAO χ^2 relative to $\Lambda_0 = 0$; the curve is flat ($\Delta\chi^2 < 0.004$ across the full range), confirming RIFT is cosmologically silent in current BAO data.